

Reconocimiento de Patrones

Versión 2026

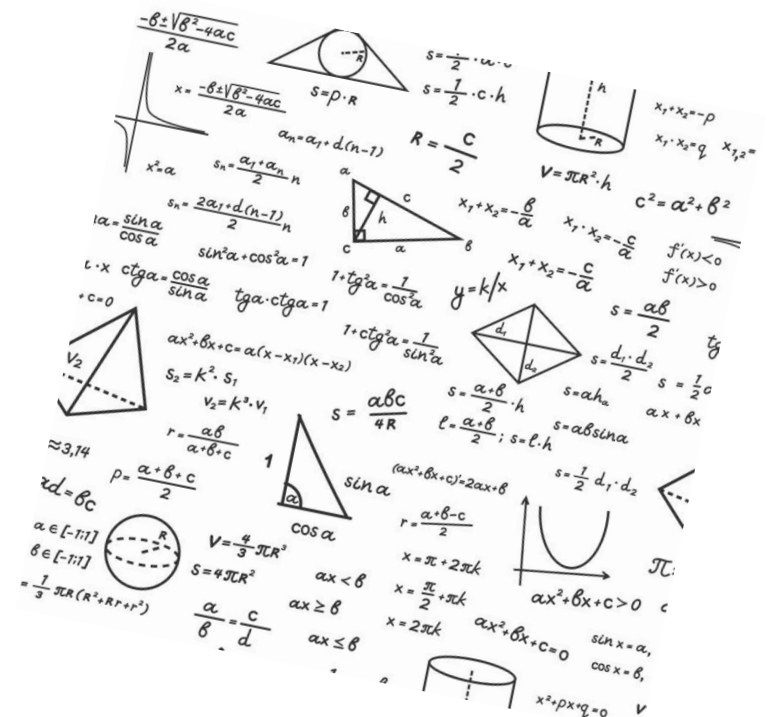


Características Geométricas

(CAPÍTULO 2 – PR02A)

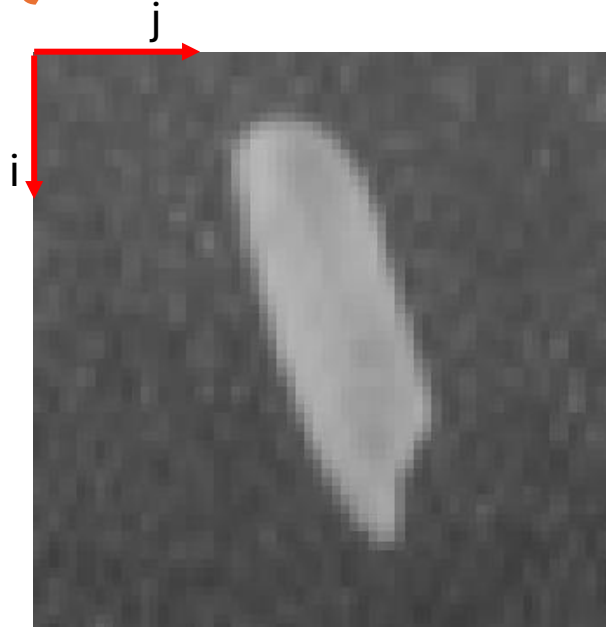
Domingo Mery 

Ciencia de la Computación
Universidad Católica – Chile
<http://domingomery.ing.uc.cl>

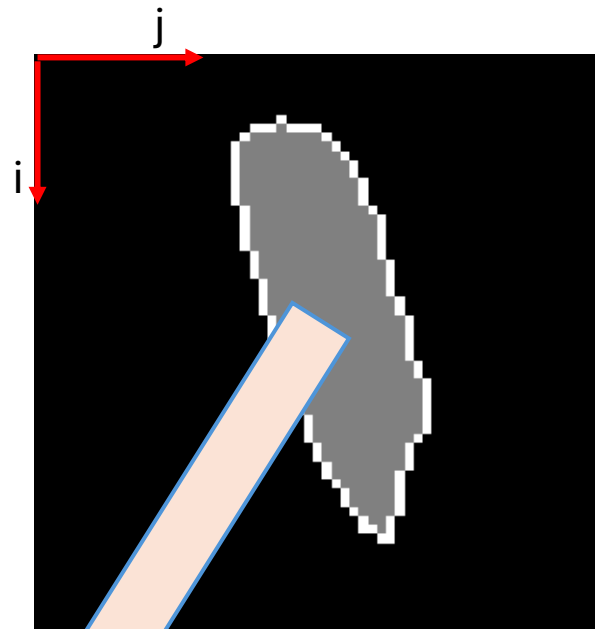


Introducción

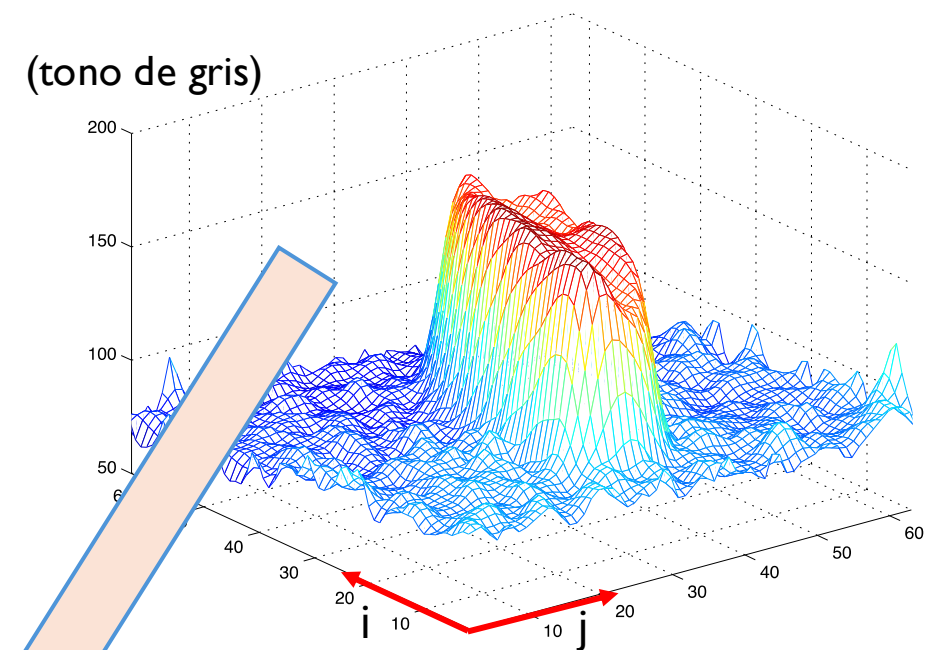
¿Qué es la Extracción de Características?



a) Imagen en escala de grises



b) Segmentación



c) Representación 3D de a)

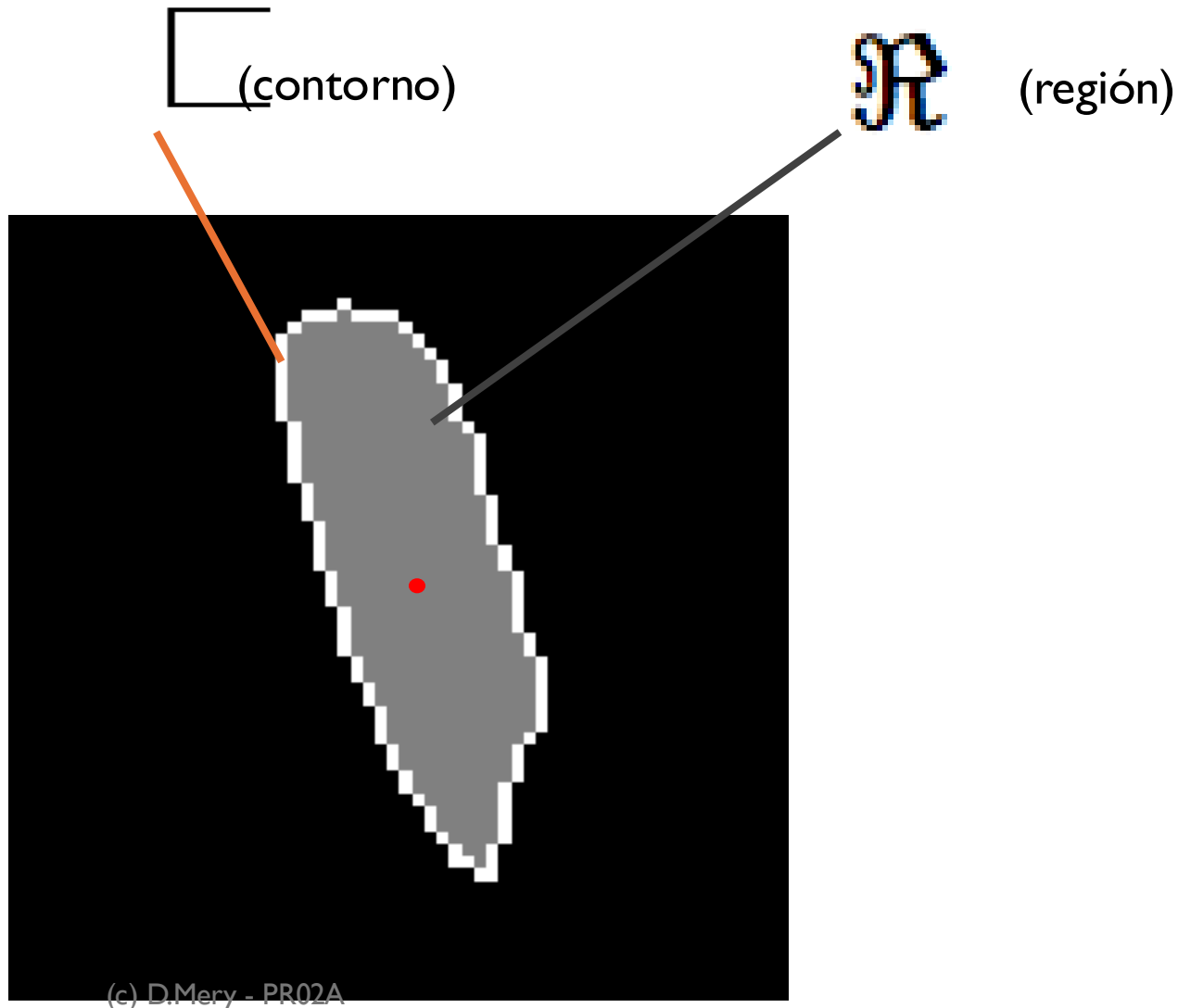
Existen dos categorías de características:

1) Las **Características Geométricas** proporcionan información sobre traslación, rotación, forma y tamaño.

2) Las **Características de Intensidad** proporcionan información sobre cómo se distribuyen los tonos de grises (o los canales de color).

Características geométricas

Características Básicas



Área y perímetro

A = # de píxeles grises

L = # de píxeles blancos

Altura y ancho de $\mathcal{R}$

$$h = i_{max} - i_{min} + 1$$

$$w = j_{max} - j_{min} + 1$$

Redondez

$$R = \frac{4 \cdot A \cdot \pi}{L^2}$$

Centro de masa

(i_m, j_m)

Ejemplo: imagen binaria 9x9 – Características Básicas

	1	2	3	4	5	6	7	8	9
1	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	1	1	0	0
5	0	0	0	0	1	1	0	0	0
6	0	0	0	1	1	0	0	0	0
7	0	0	1	1	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0

i (vertical axis), j (horizontal axis)

En la región R:

$$i = \{4, 4, 5, 5, 6, 6, 7, 7\}$$

$$j = \{3, 4, 4, 5, 5, 6, 6, 7\}$$

$$h = i_{\max} - i_{\min} + 1 = 4$$

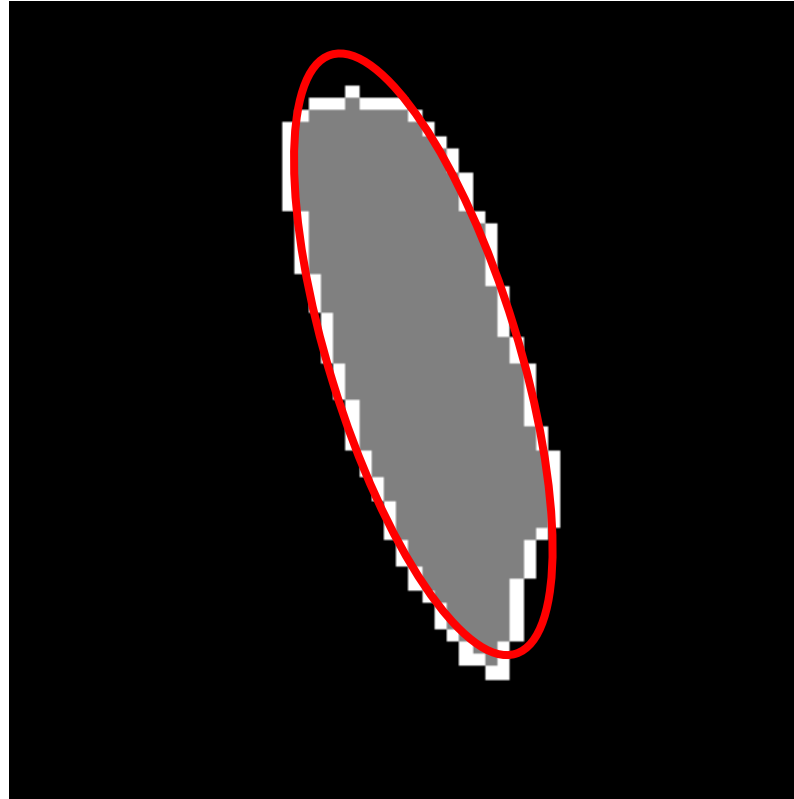
$$w = j_{\max} - j_{\min} + 1 = 5$$

$$A = 8$$

$$i_m = \frac{1}{A} \sum i = 5.5$$

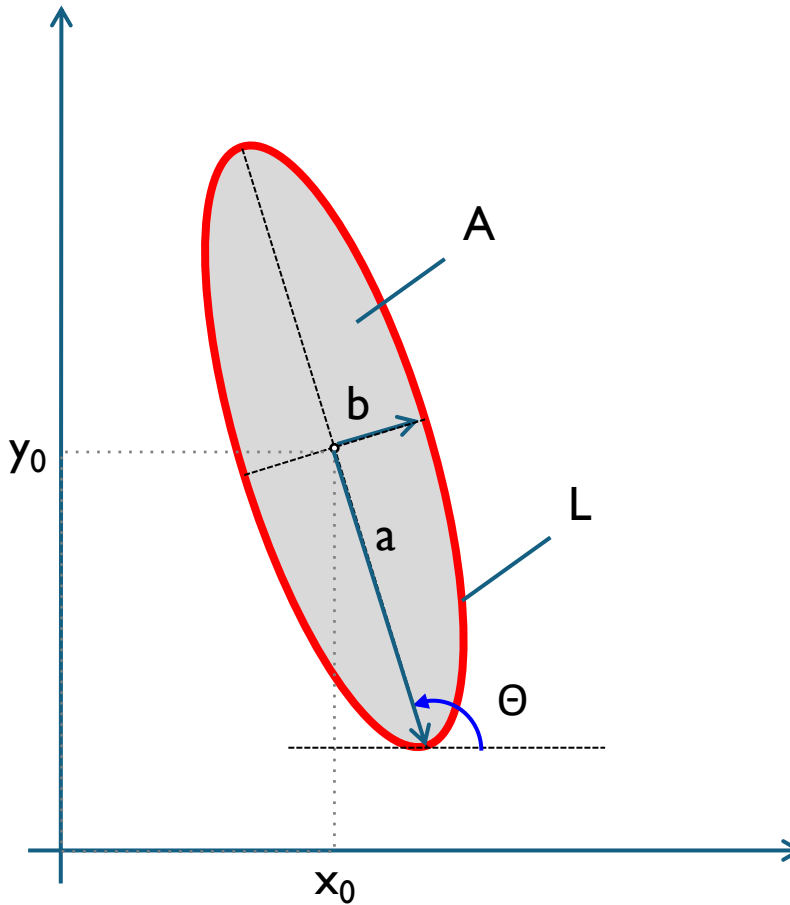
$$j_m = \frac{1}{A} \sum j = 5.0$$

Ellipses



Elipses

Eje mayor	(a)
Eje menor	(b)
Orientación	(θ)
Centro	x_0, y_0
Área	A
Perímetro	L
Excentricidad	$E = b/a$



Momentos

$$m_{rs} = \sum_{i,j \in \mathfrak{R}} i^r j^s \quad \text{for } r, s \in \mathbb{N}$$

$$\begin{array}{lll} m_{00} & \text{(centro de masa: } \bar{i}, \bar{j} \text{)} & \bar{i} = \frac{m_{10}}{m_{00}} \\ \text{(área)} & & \bar{j} = \frac{m_{01}}{m_{00}} \end{array}$$

(momentos invariantes a la traslación)

$$\mu_{rs} = \sum_{i,j \in \mathfrak{R}} (i - \bar{i})^r (j - \bar{j})^s \quad \text{for } r, s \in \mathbb{N}$$

Ejemplo: imagen binaria 9x9 - Momentos

	1	2	3	4	5	6	7	8	9
1	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	1	1	0	0
5	0	0	0	0	1	1	0	0	0
6	0	0	0	1	1	0	0	0	0
7	0	0	1	1	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0

i

En la región R:

$$i = \{4, 4, 5, 5, 6, 6, 7, 7\}$$

$$j = \{3, 4, 4, 5, 5, 6, 6, 7\}$$

$$A = m_{00} = \sum_{i,j \in R} 1 = 8$$

$$m_{10} = \sum_{i \in R} i = 44 \quad m_{01} = \sum_{j \in R} j = 40$$

$$\bar{i} = \frac{m_{10}}{m_{00}} = 5.5 \quad \bar{j} = \frac{m_{01}}{m_{00}} = 5$$

Momentos de Hu

Son invariantes al:

- tamaño (escala),
- rotación y
- traslación

$$\phi_1 = \eta_{20} + \eta_{02}$$

$$\phi_2 = (\eta_{20} - \eta_{02})^2 + 4\eta_{11}^2$$

$$\phi_3 = (\eta_{30} - 3\eta_{12})^2 + (3\eta_{21} - \eta_{03})^2$$

$$\phi_4 = (\eta_{30} + \eta_{12})^2 + (\eta_{21} + \eta_{03})^2$$

$$\phi_5 = (\eta_{30} - 3\eta_{12})(\eta_{30} + \eta_{12})[(\eta_{30} + \eta_{12})^2 - 3(\eta_{21} + \eta_{03})^2] + \\ (3\eta_{21} - \eta_{03})(\eta_{21} + \eta_{03})[3(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2]$$

$$\phi_6 = (\eta_{20} - \eta_{02})[(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2] + \\ 4\eta_{11}(\eta_{30} + \eta_{12})(\eta_{21} + \eta_{03})$$

$$\phi_7 = (3\eta_{21} - \eta_{03})(\eta_{30} + \eta_{12})[(\eta_{30} + \eta_{12})^2 - 3(\eta_{21} + \eta_{03})^2] - \\ (\eta_{30} - 3\eta_{12})(\eta_{21} + \eta_{03})[3(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2]$$

con

$$\eta_{rs} = \frac{\mu_{rs}}{\mu_{00}^t} \quad t = \frac{r+s}{2} + 1.$$

Ejemplo: imagen binaria 9x9 – Momentos de Hu

	1	2	3	4	5	6	7	8	9
1	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	1	1	0	0
5	0	0	0	0	1	1	0	0	0
6	0	0	0	1	1	0	0	0	0
7	0	0	1	1	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0

i (vertical axis), j (horizontal axis)

En la región R:

$$i = \{4, 4, 5, 5, 6, 6, 7, 7\}$$

$$j = \{3, 4, 4, 5, 5, 6, 6, 7\}$$

$$\eta_{20} = \frac{\mu_{20}}{\mu_{00}^2} = 0.1562$$

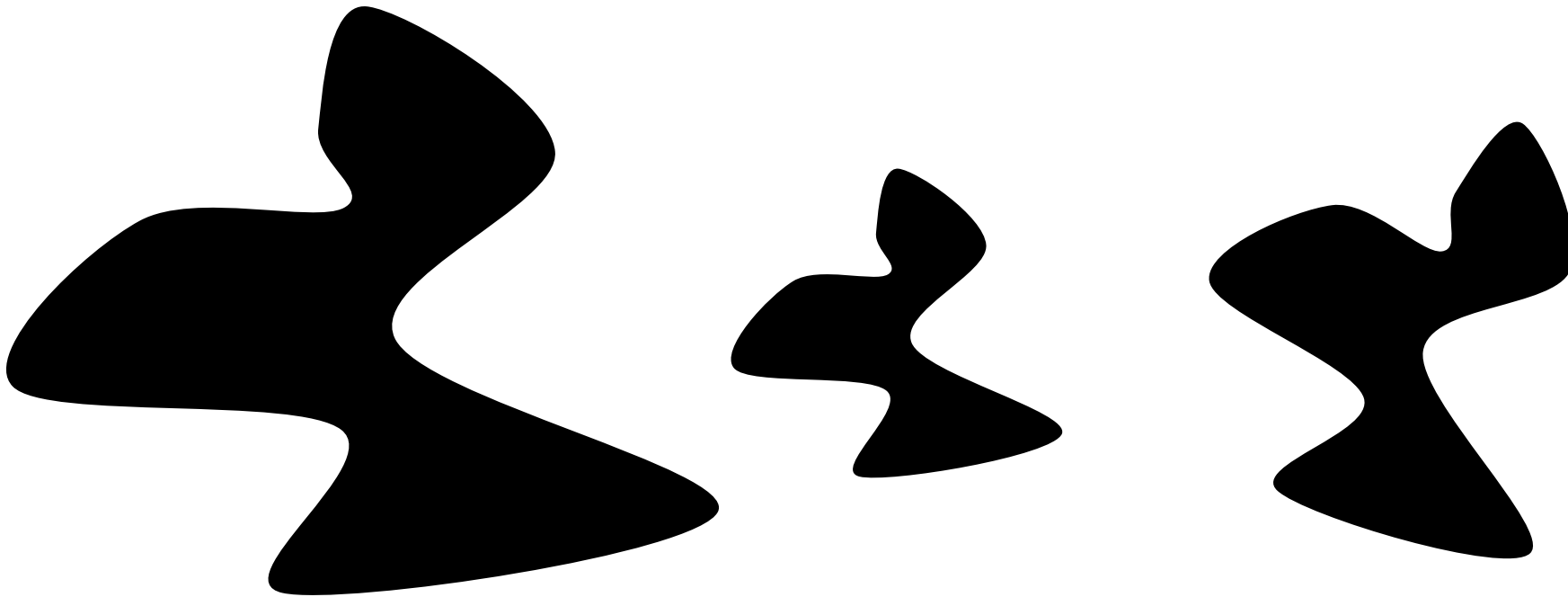
$$\eta_{02} = \frac{\mu_{02}}{\mu_{00}^2} = 0.1875$$

$$\phi_1 = \eta_{20} + \eta_{02} = 0.3438$$

Momentos de Hu

Son invariantes al:

- tamaño (escala),
- rotación y
- traslación



Los valores $\Phi_1, \Phi_2, \dots, \Phi_7$ son similares

Momentos de Hu

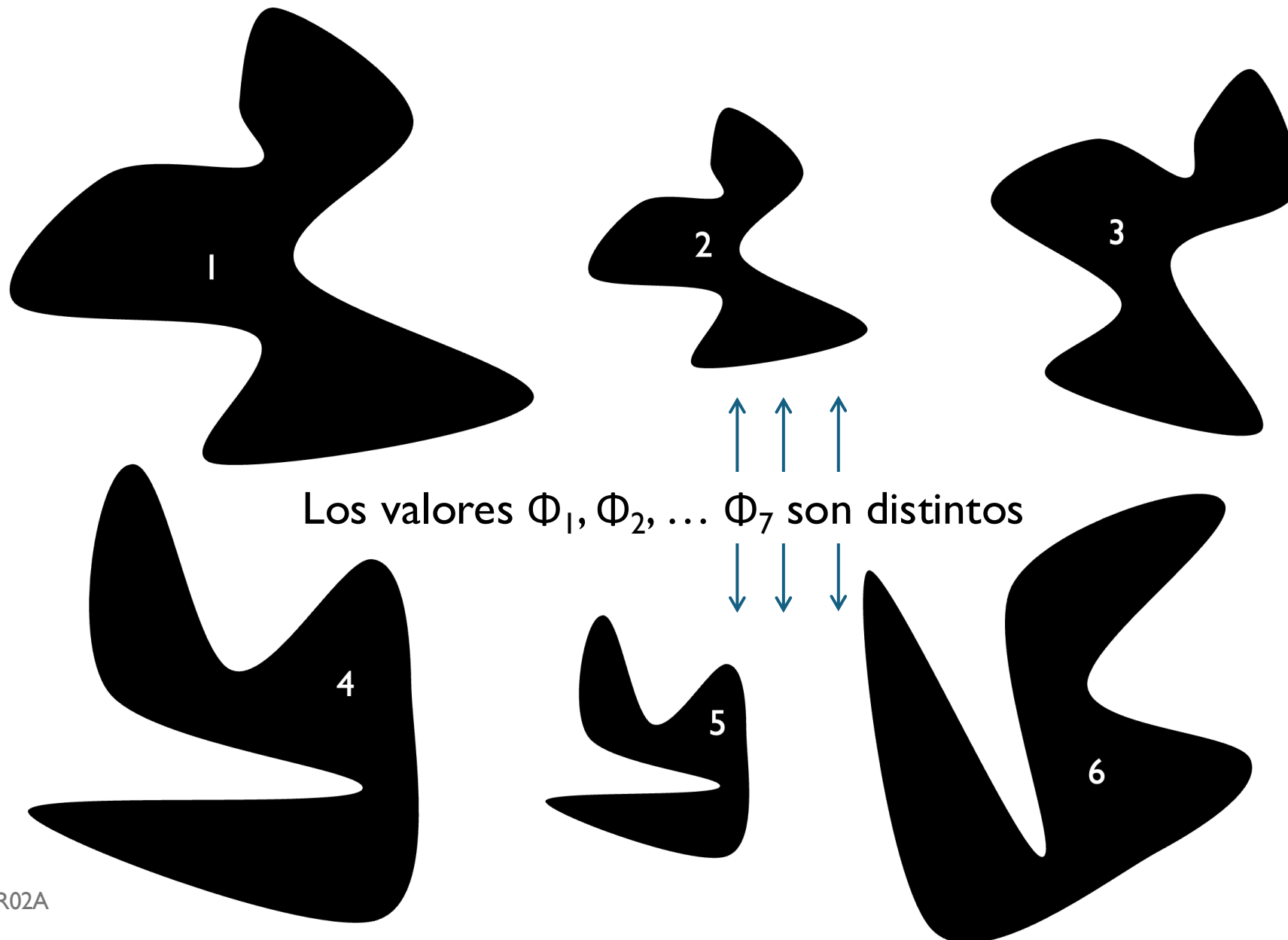
Son invariantes al:

- tamaño (escala),
- rotación y
- traslación

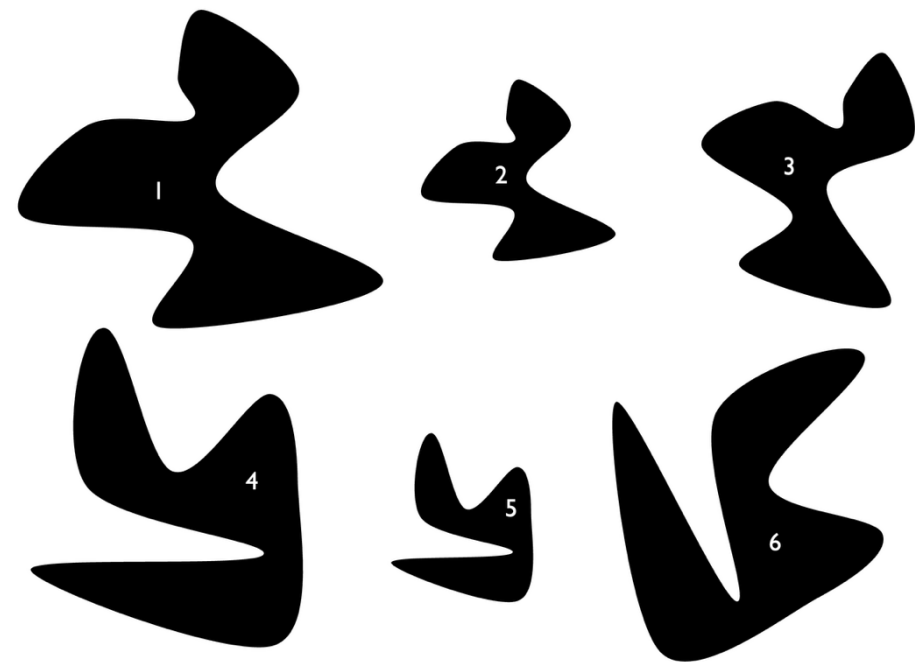
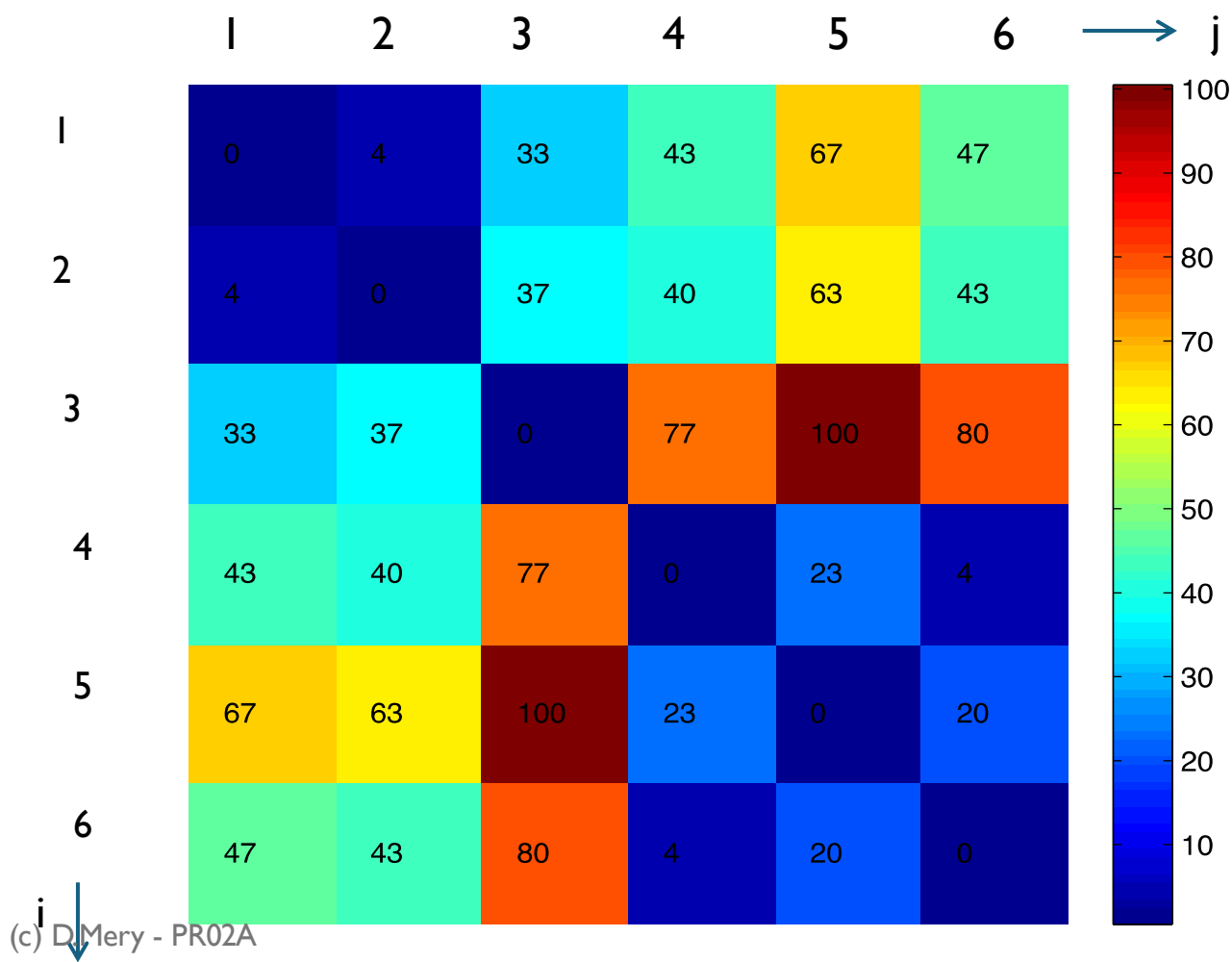


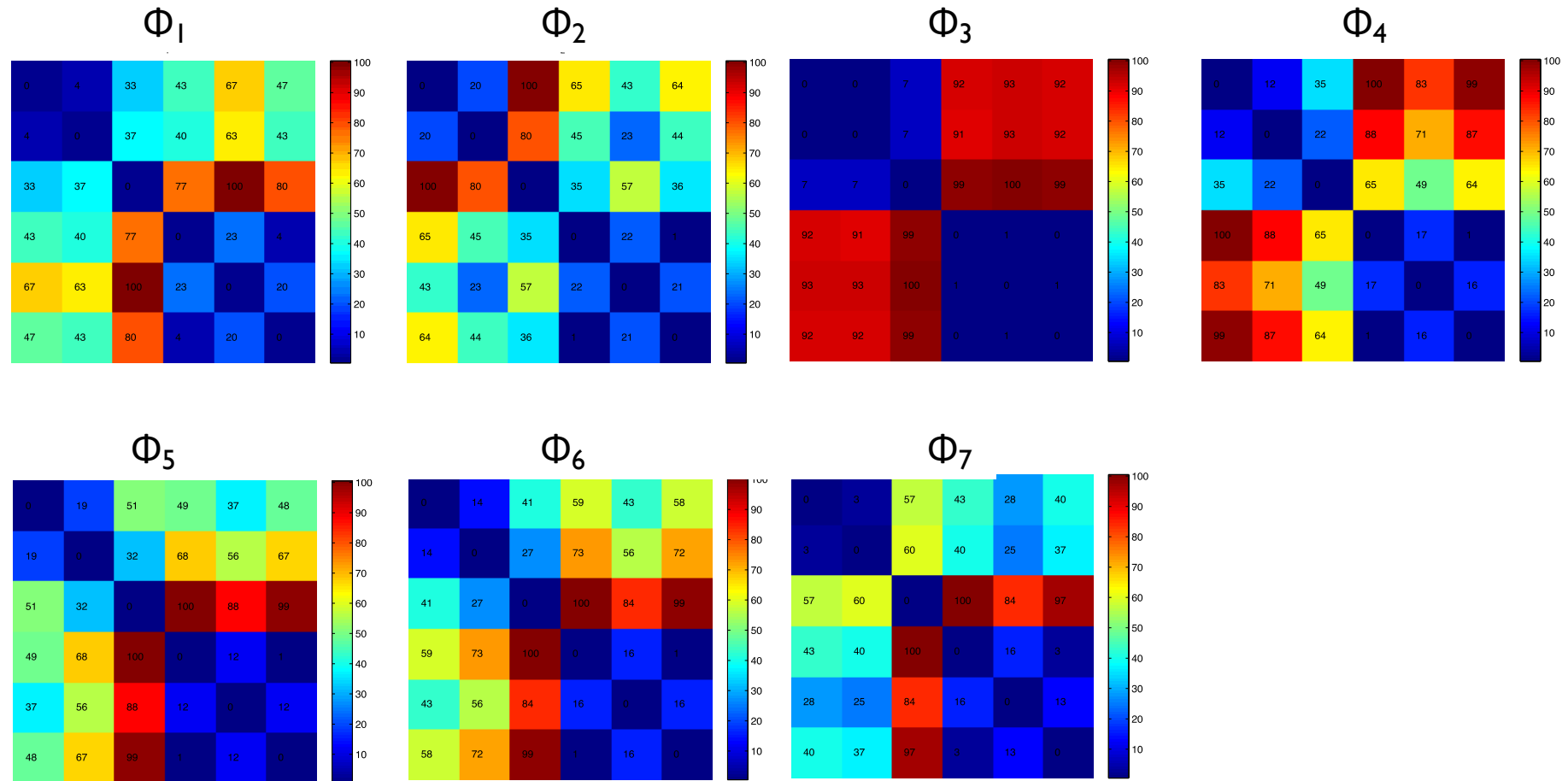
Los valores $\Phi_1, \Phi_2, \dots, \Phi_7$ son similares

Momentos de Hu

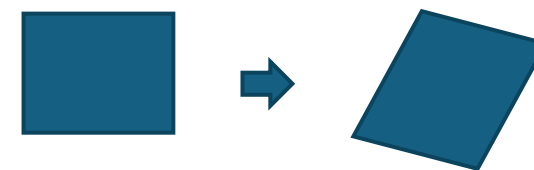


Diferencia entre Φ_i de la región i y Φ_i de la región j





Momentos de Flusser



Invariantes a Escala, Rotación y
Traslación

+

Invariante a la transformada afín:
líneas paralelas se transforman
como líneas paralelas

$$I_1 = \frac{\mu_{20}\mu_{02} - \mu_{11}^2}{\mu_{00}^4}$$

$$I_2 = \frac{\mu_{30}^2\mu_{03}^2 - 6\mu_{30}\mu_{21}\mu_{12}\mu_{03} + 4\mu_{30}\mu_{12}^3 + 4\mu_{21}^3\mu_{03} - 3\mu_{21}^2\mu_{12}^2}{\mu_{00}^{10}}$$

$$I_3 = \frac{\mu_{20}(\mu_{21}\mu_{03} - \mu_{12}^2) - \mu_{11}(\mu_{30}\mu_{03} - \mu_{21}\mu_{12}) + \mu_{02}(\mu_{30}\mu_{12} - \mu_{21}^2)}{\mu_{00}^7}$$

$$I_4 = \frac{(\mu_{20}^3\mu_{03}^2 - 6\mu_{20}^2\mu_{11}\mu_{12}\mu_{03} - 6\mu_{20}^2\mu_{02}\mu_{21}\mu_{03} + 9\mu_{20}^2\mu_{02}\mu_{12}^2 + 12\mu_{20}\mu_{11}^2\mu_{21}\mu_{03} + 6\mu_{20}\mu_{11}\mu_{02}\mu_{30}\mu_{03} - 18\mu_{20}\mu_{11}\mu_{02}\mu_{21}\mu_{12} - 8\mu_{11}^3\mu_{30}\mu_{03} - 6\mu_{20}\mu_{02}^2\mu_{30}\mu_{12} + 9\mu_{20}\mu_{02}^2\mu_{21} + 12\mu_{11}^2\mu_{02}\mu_{30}\mu_{12} - 6\mu_{11}\mu_{02}^2\mu_{30}\mu_{21} + \mu_{02}^3\mu_{30}^2)/\mu_{00}^{11}}$$

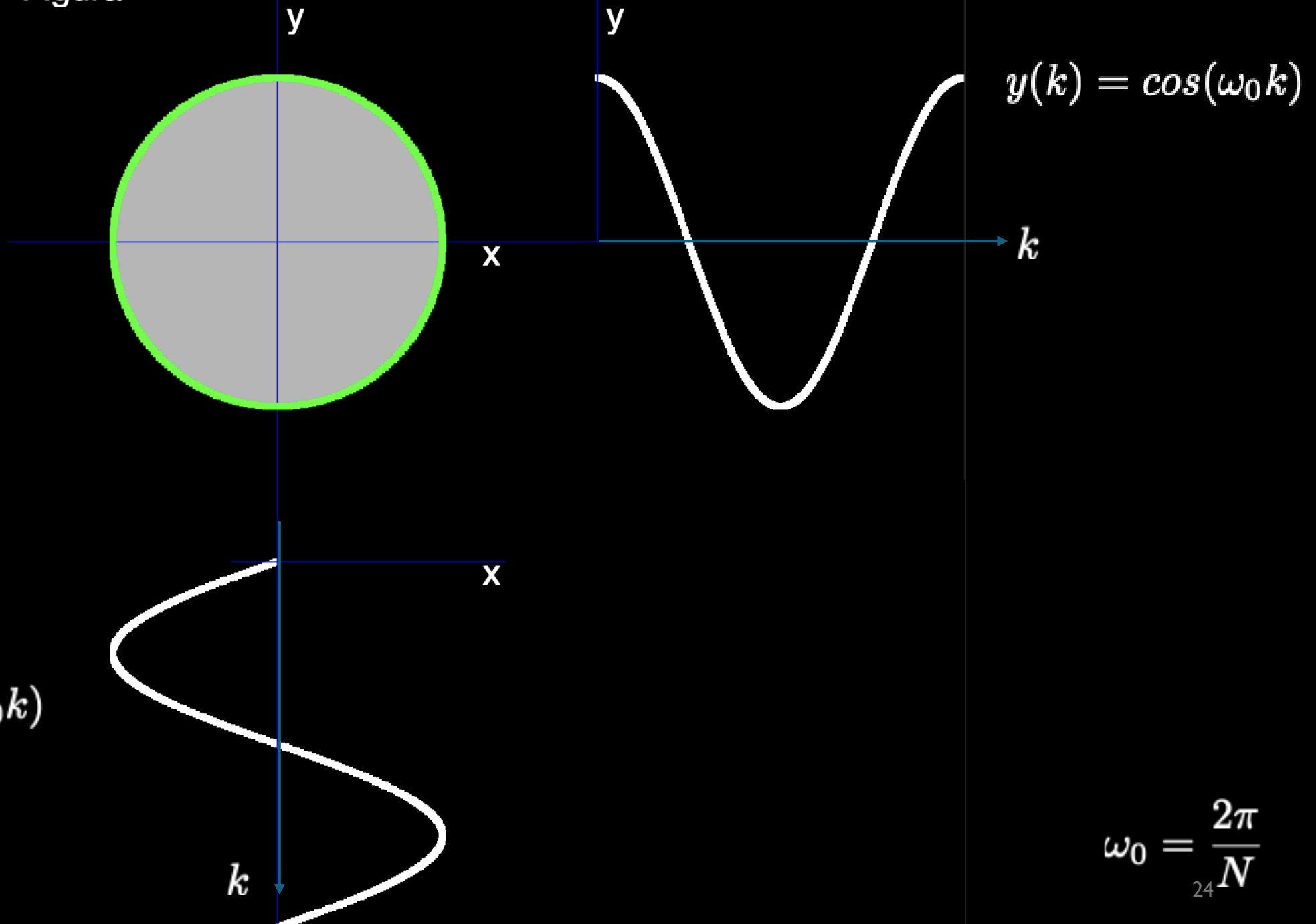
Flusser, J., & Suk, T. (1993). [Pattern recognition by affine moment invariants](#). Pattern Recognition, 26(1), 167-174.

Descriptores de Fourier

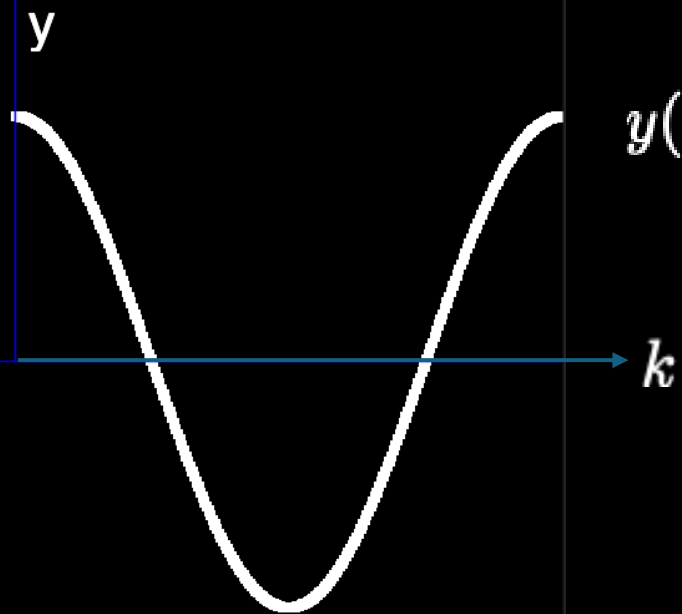
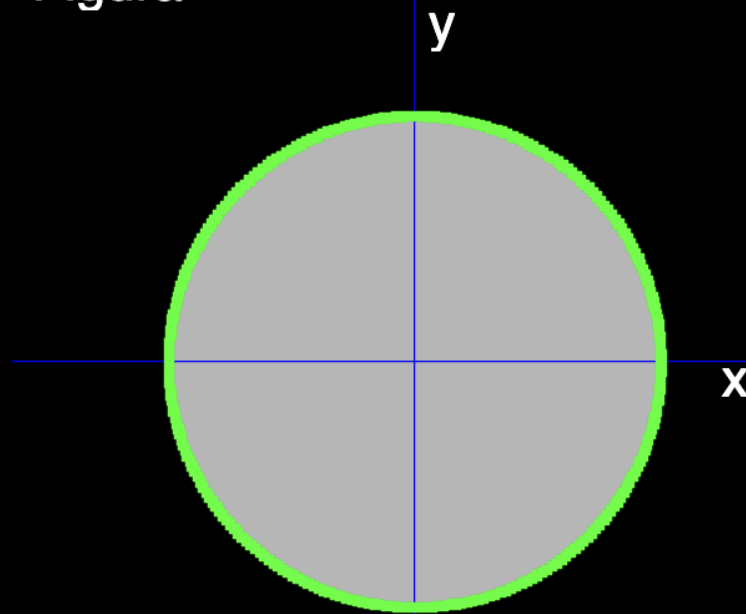
Descriptores de Fourier

Idea Principal

Figura

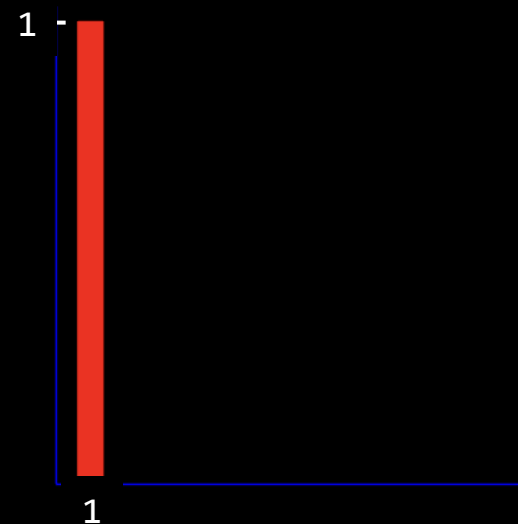


Figura

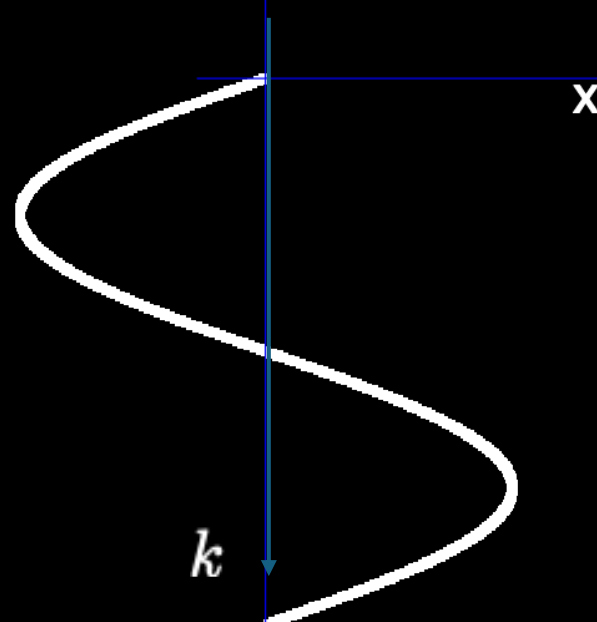


$$y(k) = \cos(\omega_0 k)$$

Frecuencias



$$x(k) = \sin(\omega_0 k)$$



$$\omega_0 = \frac{2\pi}{N}$$

$$y(k) = \cos(\omega_0 k) + 0.05 \cos(14\omega_0 k)$$

k

1 -

$$x(k) = \sin(\omega_0 k) + 0.05 \sin(14\omega_0 k)$$

k

0.05

$$\omega_0 = \frac{2\pi}{N}$$

14

$$\begin{aligned}
 x(k) = & \sin(\omega_0 k) \\
 & + 0.02 \sin(4\omega_0 k) \\
 & + 0.04 \sin(5\omega_0 k) \\
 & + 0.07 \sin(11\omega_0 k) \\
 & + 0.05 \sin(14\omega_0 k)
 \end{aligned}$$

$$\begin{aligned}
 y(k) = & \cos(\omega_0 k) \\
 & + 0.02 \cos(4\omega_0 k) \\
 & + 0.04 \cos(5\omega_0 k) \\
 & + 0.07 \cos(11\omega_0 k) \\
 & + 0.05 \cos(14\omega_0 k)
 \end{aligned}$$

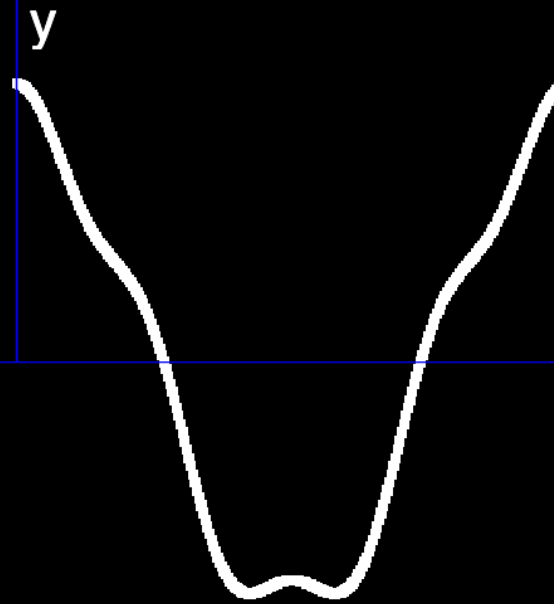
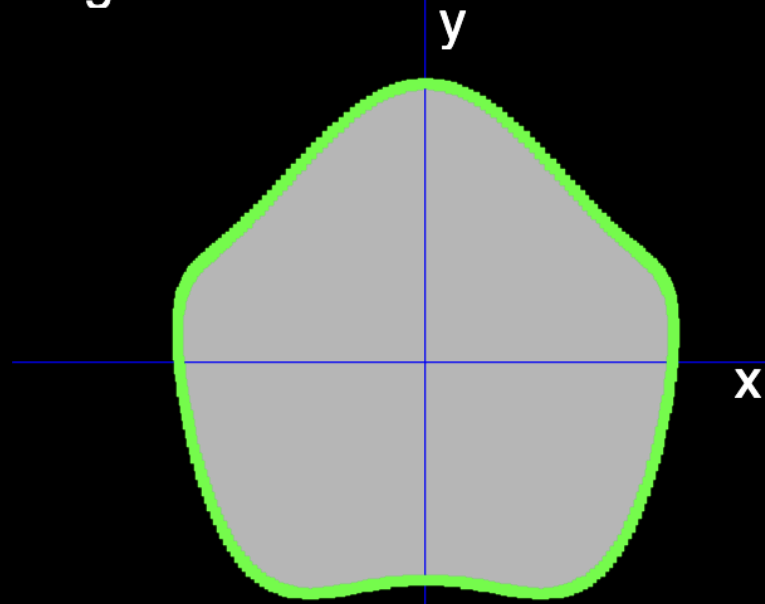
k

1 -

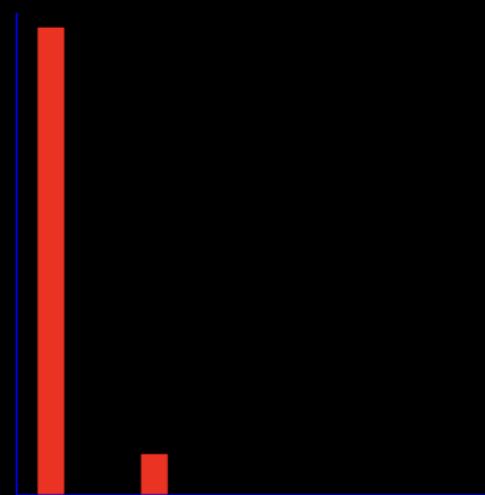
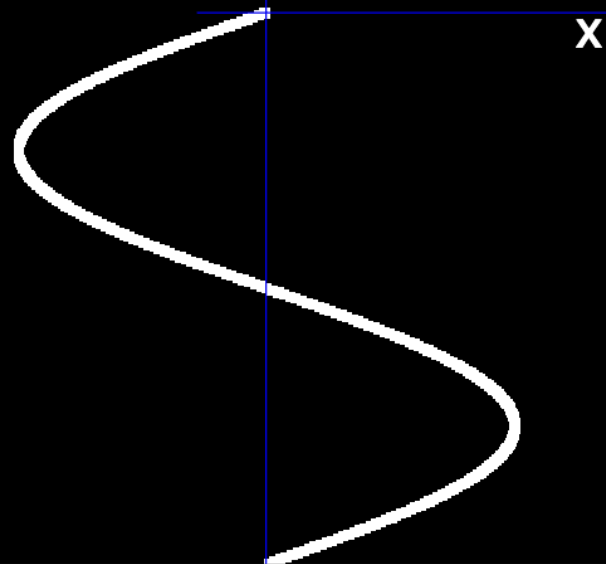
k

$$\omega_0 = \frac{2\pi}{N}$$

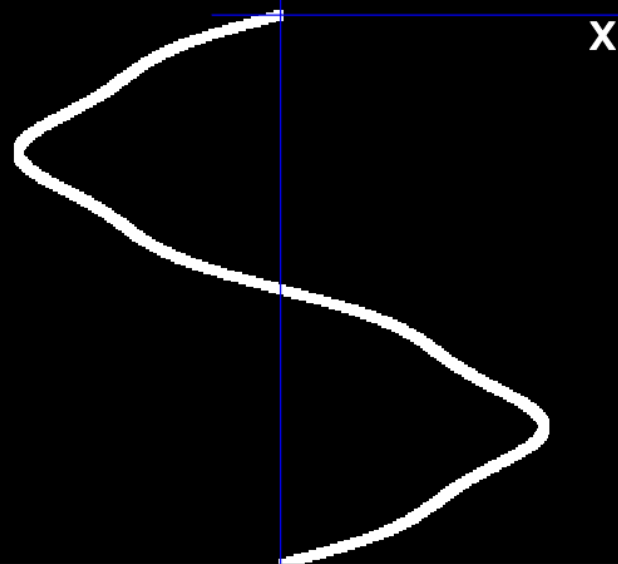
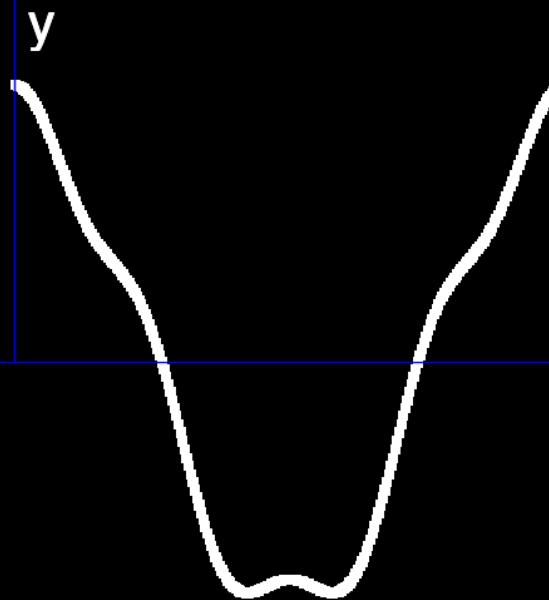
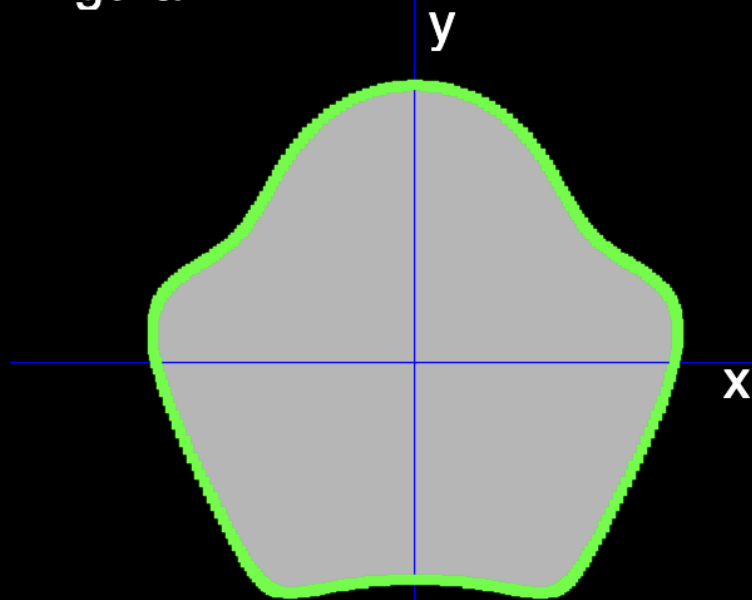
Figura



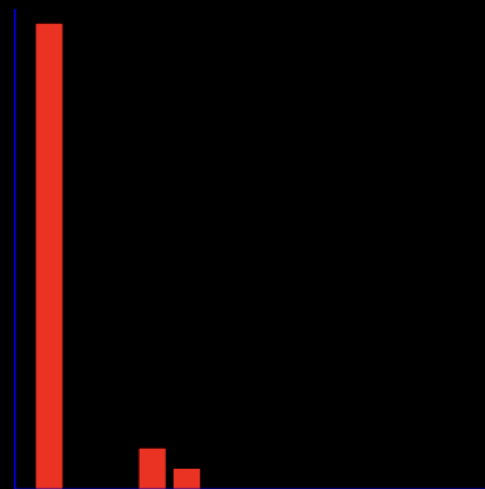
Frecuencias



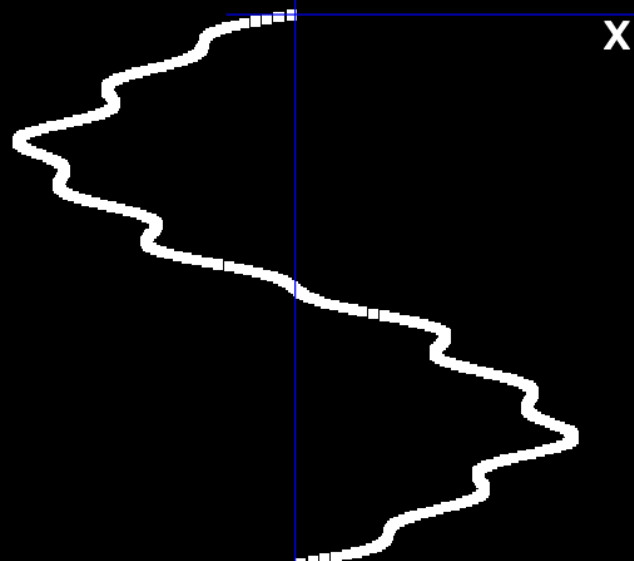
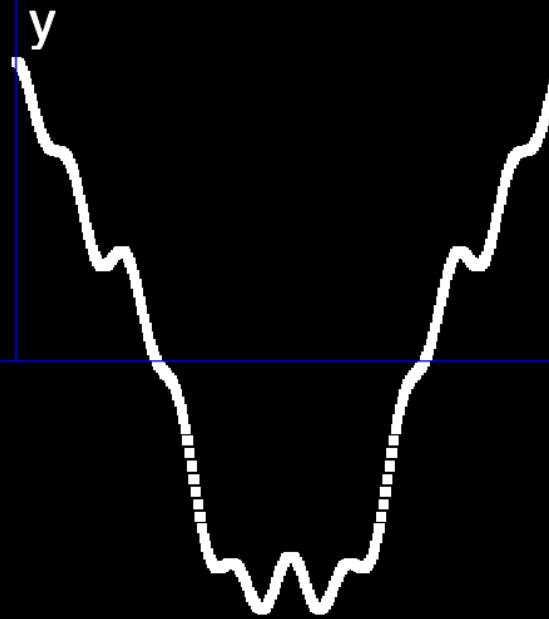
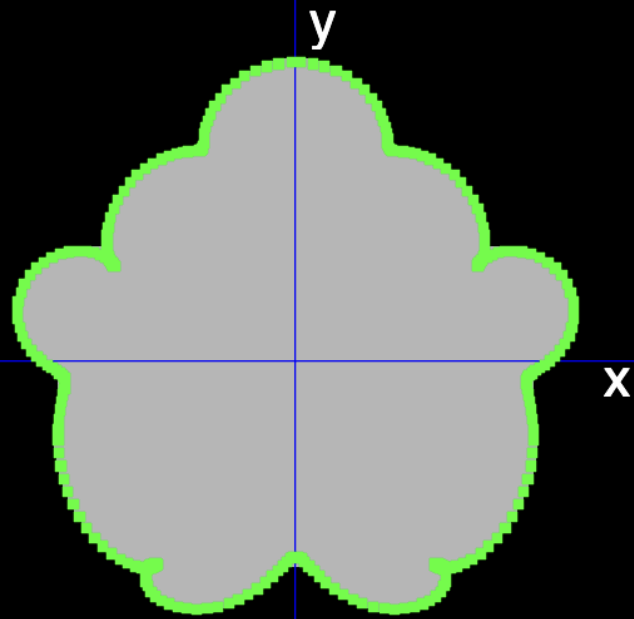
Figura



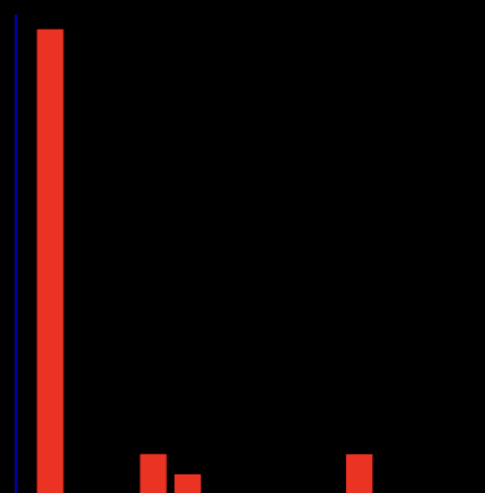
Frecuencias



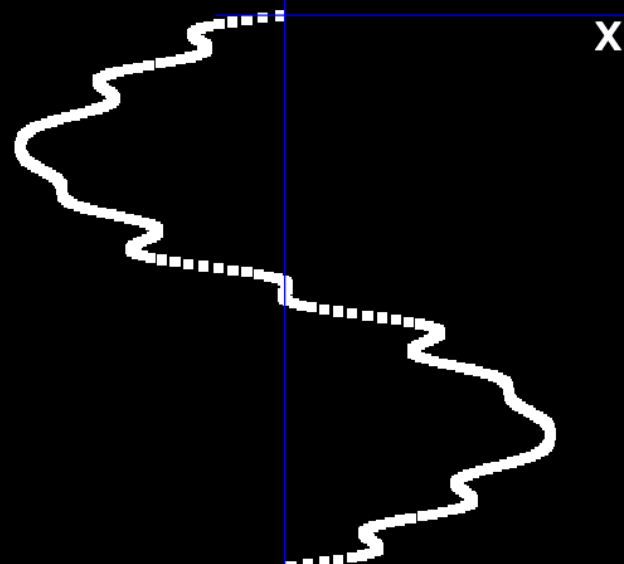
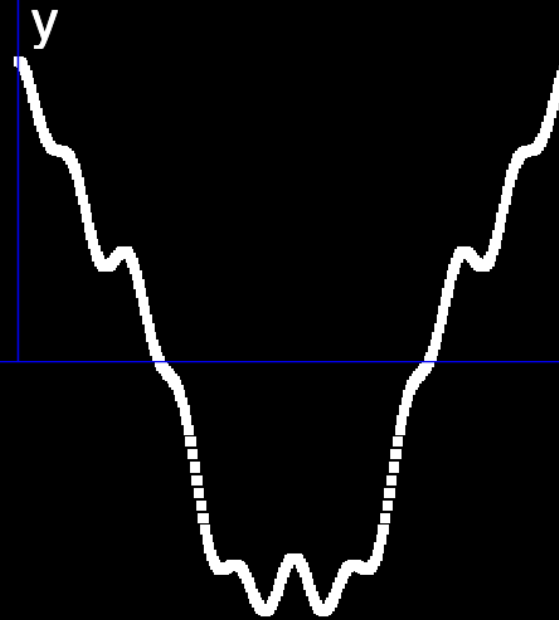
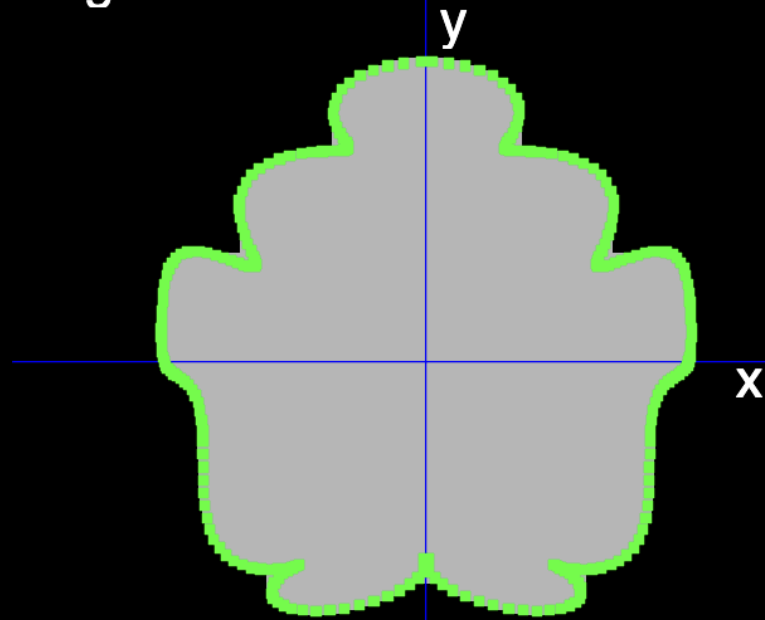
Figura



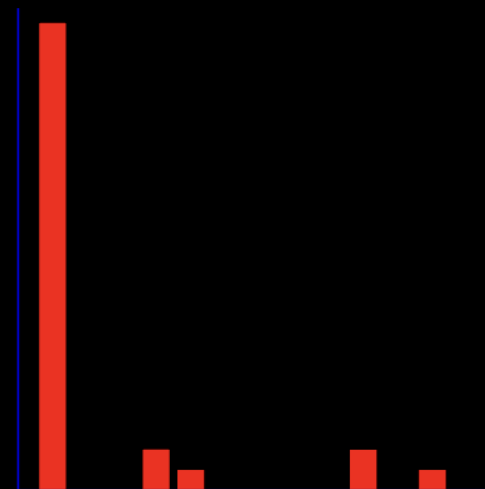
Frecuencias



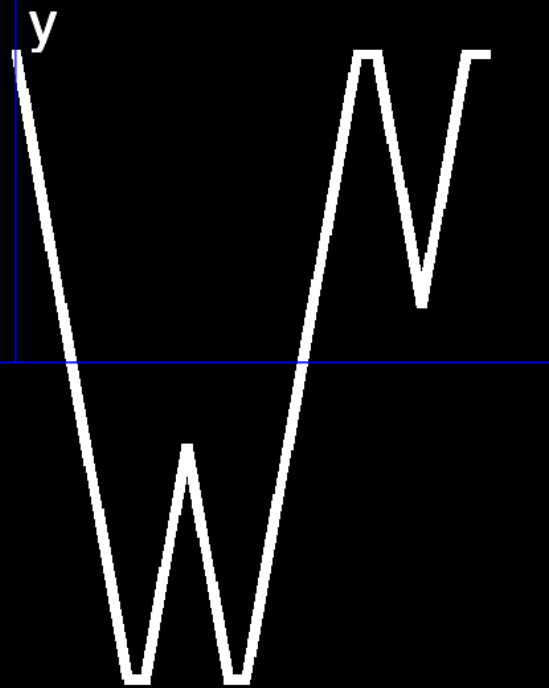
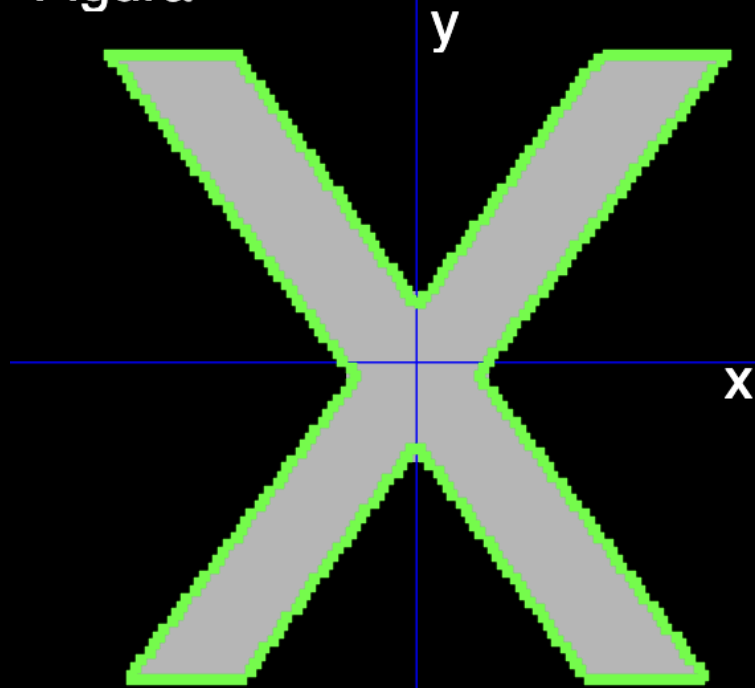
Figura



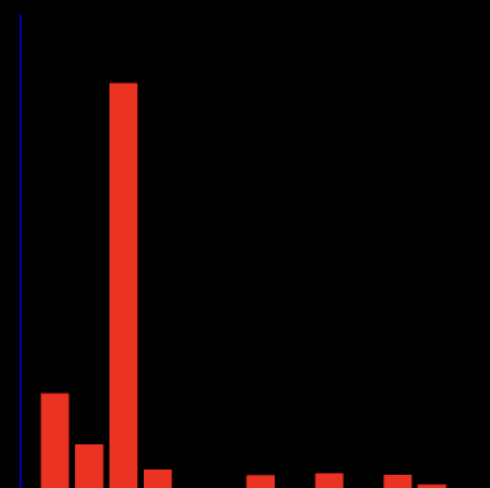
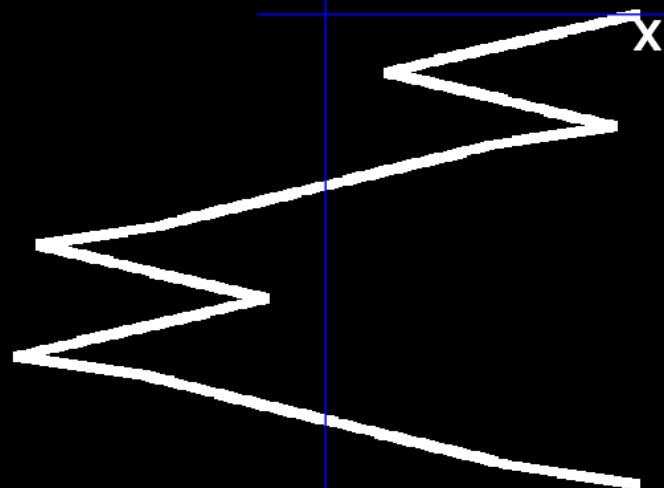
Frecuencias



Figura

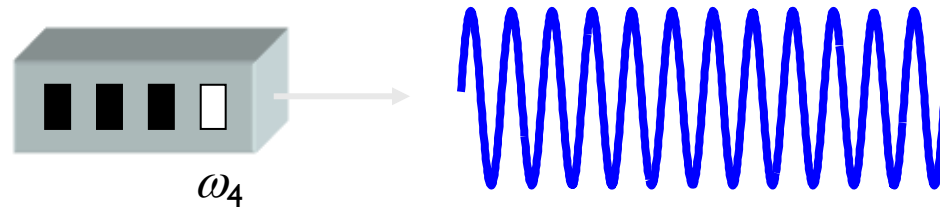
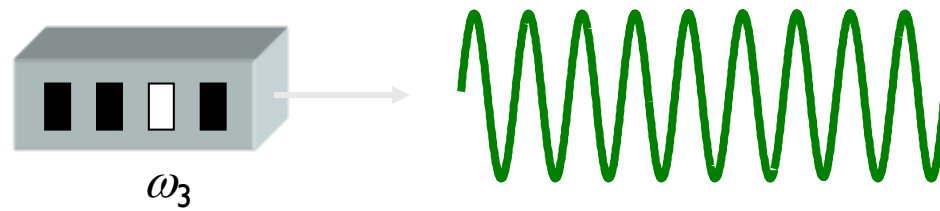
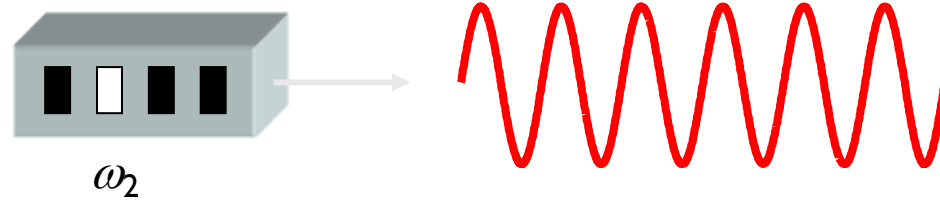
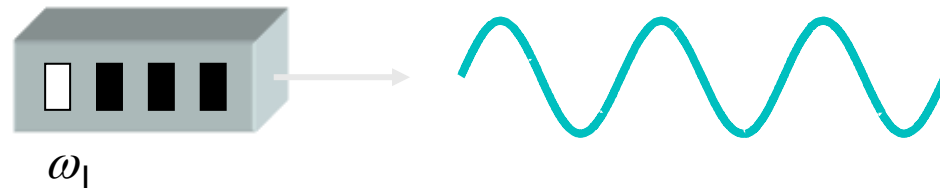


Frecuencias

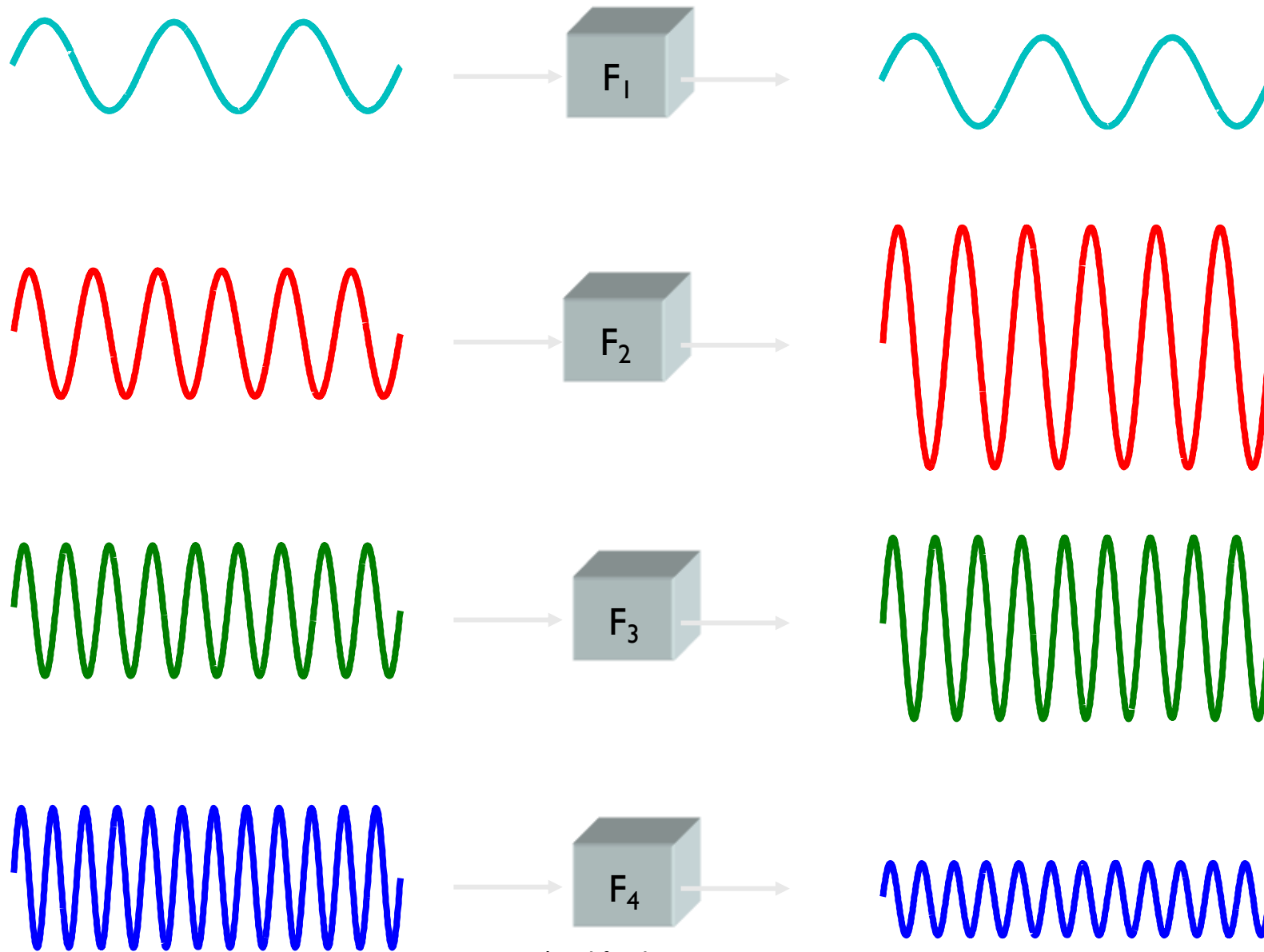


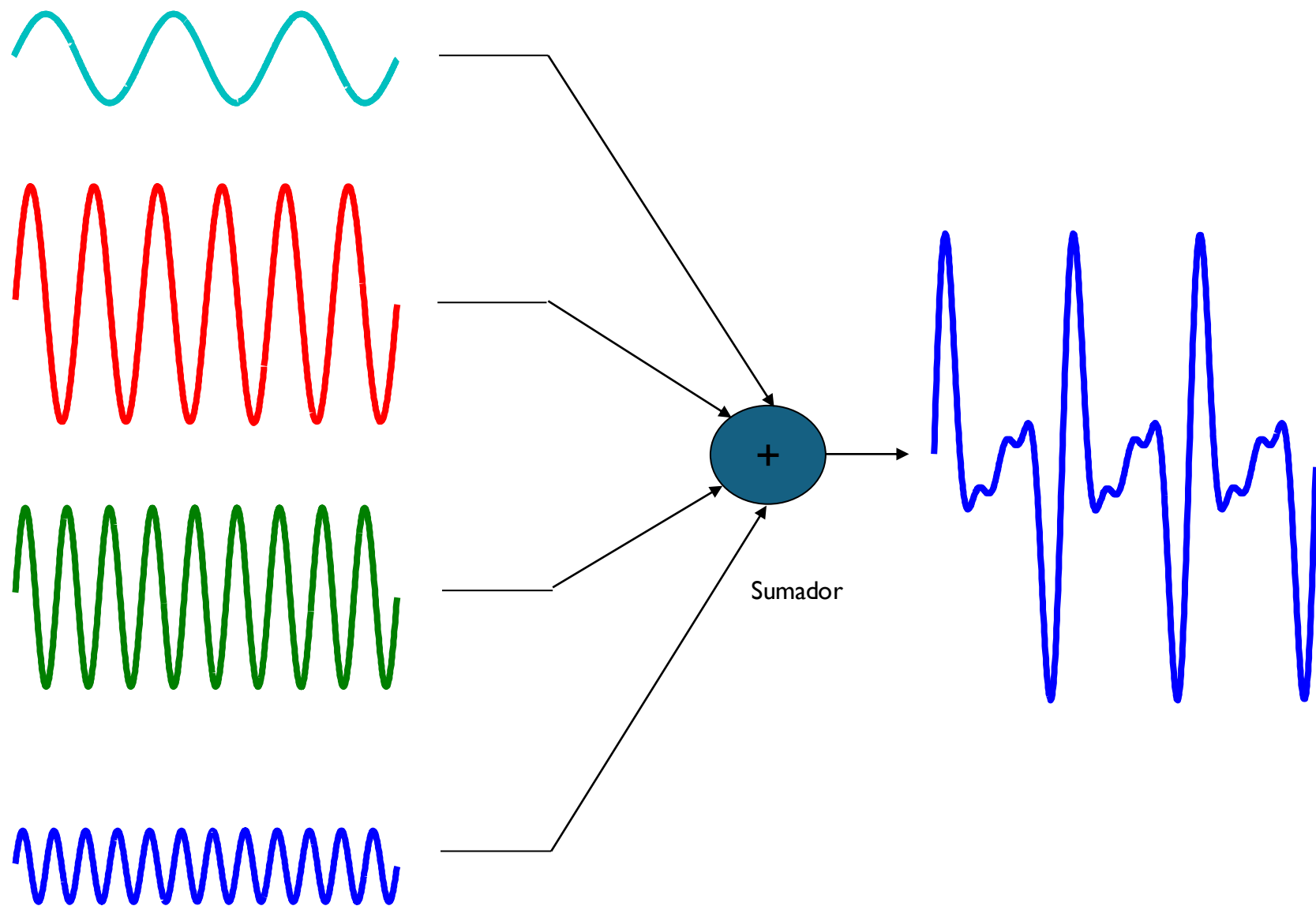
Descriptores de Fourier

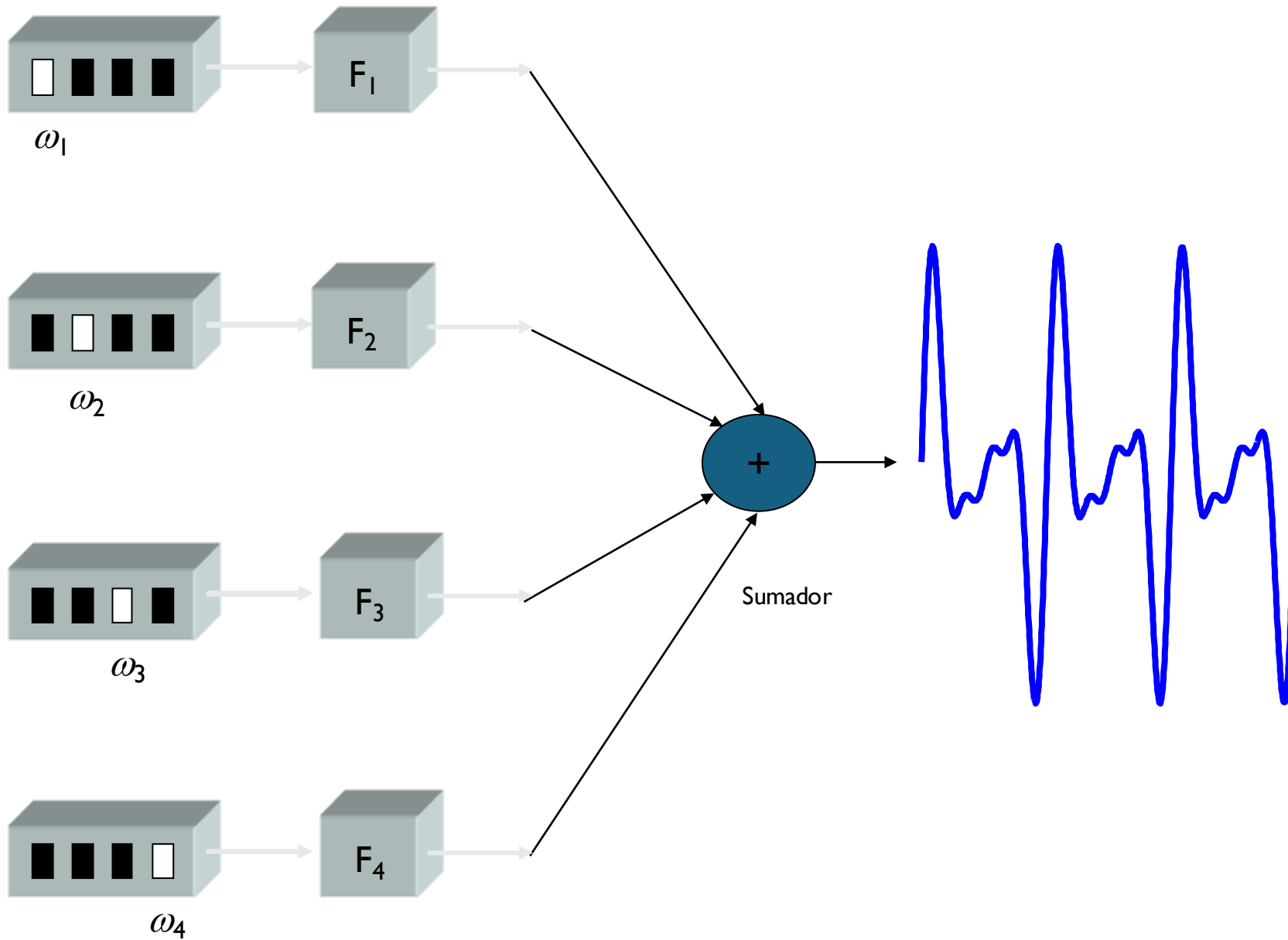
Explicación Simple de la
Transformada de Fourier

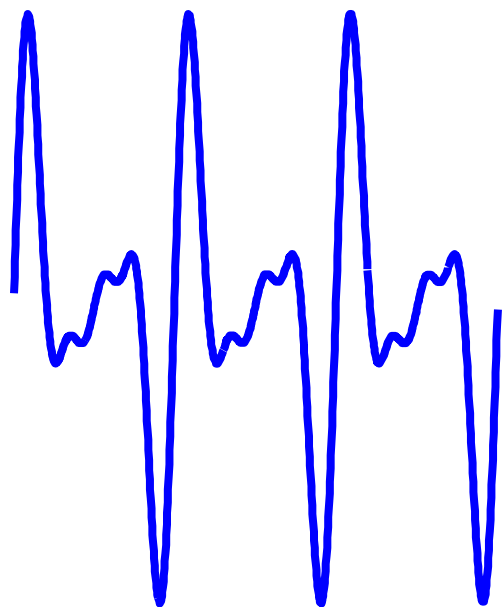


Osciladores sinusoidales



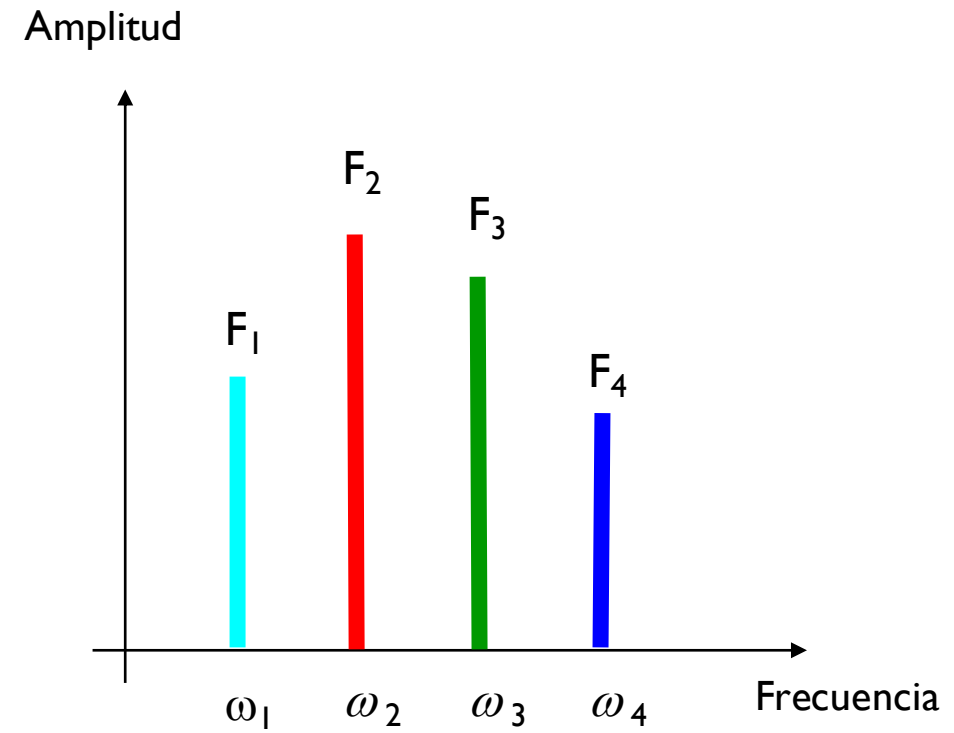


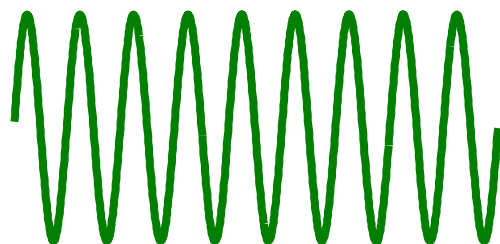
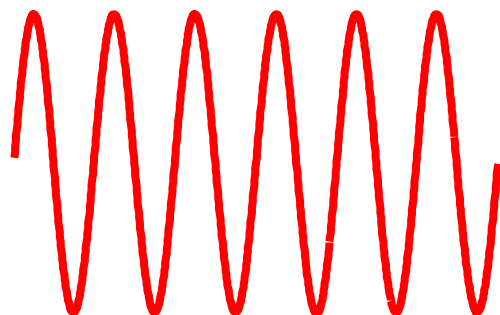
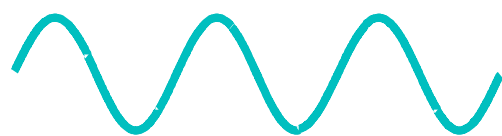




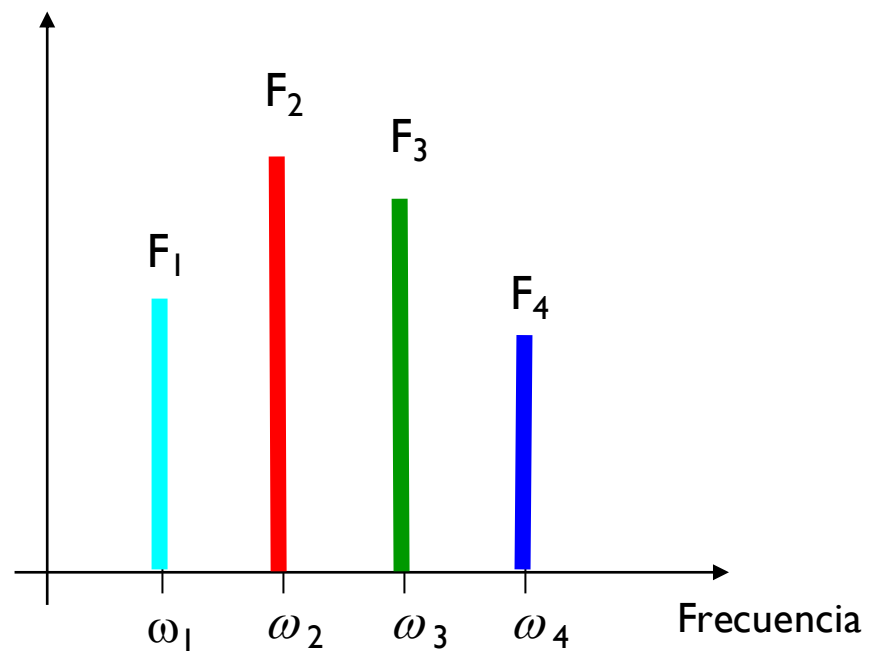
Frecuencia	Amplitud
ω_1	F_1
ω_2	F_2
ω_3	F_3
ω_4	F_4

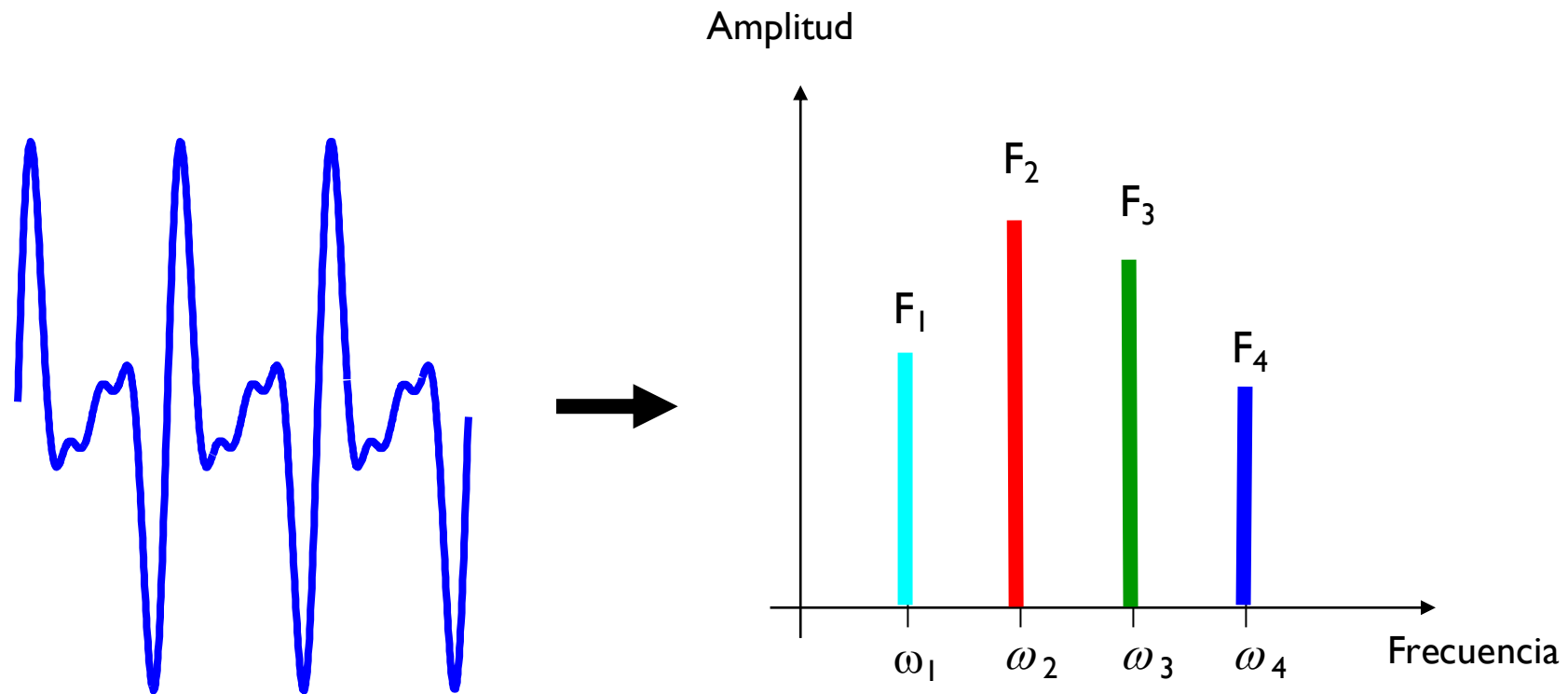
Frecuencia	Amplitud
ω_1	F_1
ω_2	F_2
ω_3	F_3
ω_4	F_4

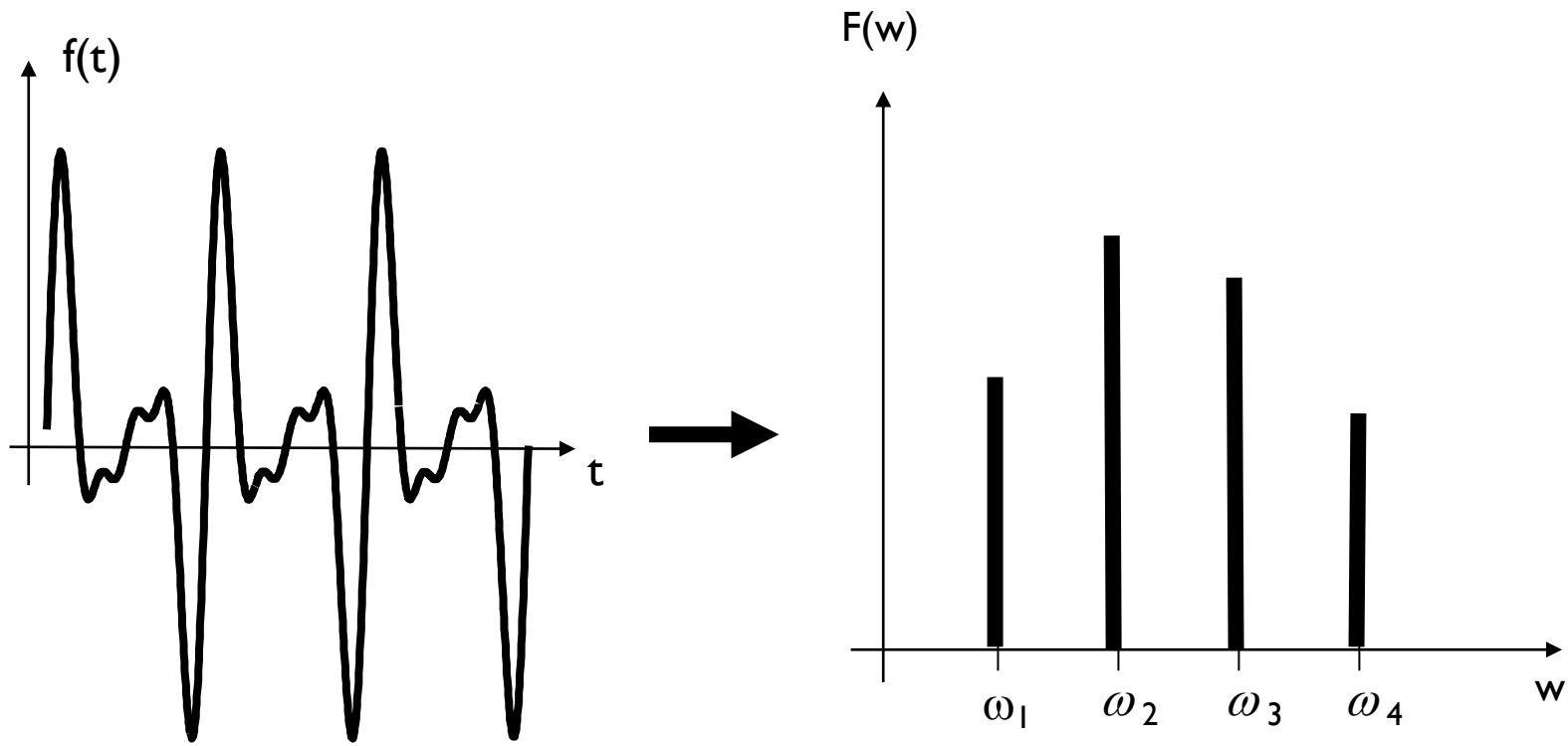




Amplitud







Señal $f(t)$

Espectro de la señal $F(w)$

$f(t)$ ● — ○ $F(w)$

Descriptores de Fourier

Matemática de la
Transformada de Fourier

Transformada de Fourier

CONCEPTOS BÁSICOS

$$f(t) \longleftrightarrow F(\omega)$$

Función en el
dominio
del tiempo

Función en el
dominio
de la frecuencia

Transformada de Fourier

CONTINUA

$$F(\omega) = \int_{-\infty}^{+\infty} f(t) e^{-j2\pi\omega t} dt$$



$$f(t) = \int_{-\infty}^{+\infty} F(\omega) e^{+j2\pi\omega t} d\omega$$

$$j = \sqrt{-1}$$

Transformada de Fourier

DISCRETA

$$F(u) = \frac{1}{N} \sum_{x=0}^{N-1} f(x) e^{-j \frac{2\pi}{N} ux}$$



$$f(x) = \sum_{u=0}^{N-1} F(u) e^{+j \frac{2\pi}{N} ux}$$

$$j = \sqrt{-1}$$

¿Cómo usar Fourier para describir la forma de una figura geométrica?

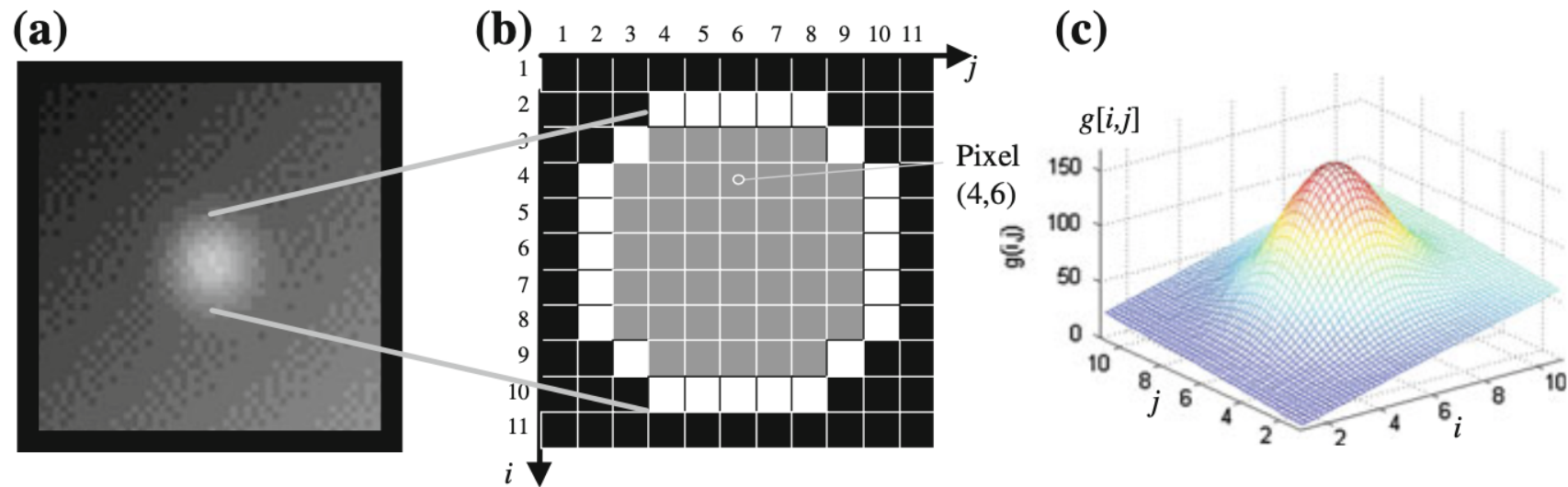


Fig. 5.1 Example of a region: **a** X-ray image, **b** segmented region (*gray* pixels), **c** 3D representation of the gray values

5.2.3 *Fourier Descriptors*

Shape information—invariant to scale, orientation and position—can be measured using *Fourier descriptors* [5–7]. The coordinates of the pixels of the boundary are arranged as a complex number $i_k + j \cdot j_k$, with $j = \sqrt{-1}$ and $k = 0, \dots, L - 1$, where L is the perimeter of the region, and pixel k and $k + 1$ are connected. The complex boundary function can be considered as a periodical signal of period L . The Discrete Fourier Transformation [8] gives a characterization of the shape of the region. The Fourier coefficients are defined by:

$$F_n = \sum_{k=0}^{L-1} (i_k + j \cdot j_k) e^{-j \frac{2\pi kn}{L}} \quad \text{for } n = 0, \dots, L - 1. \quad (5.10)$$

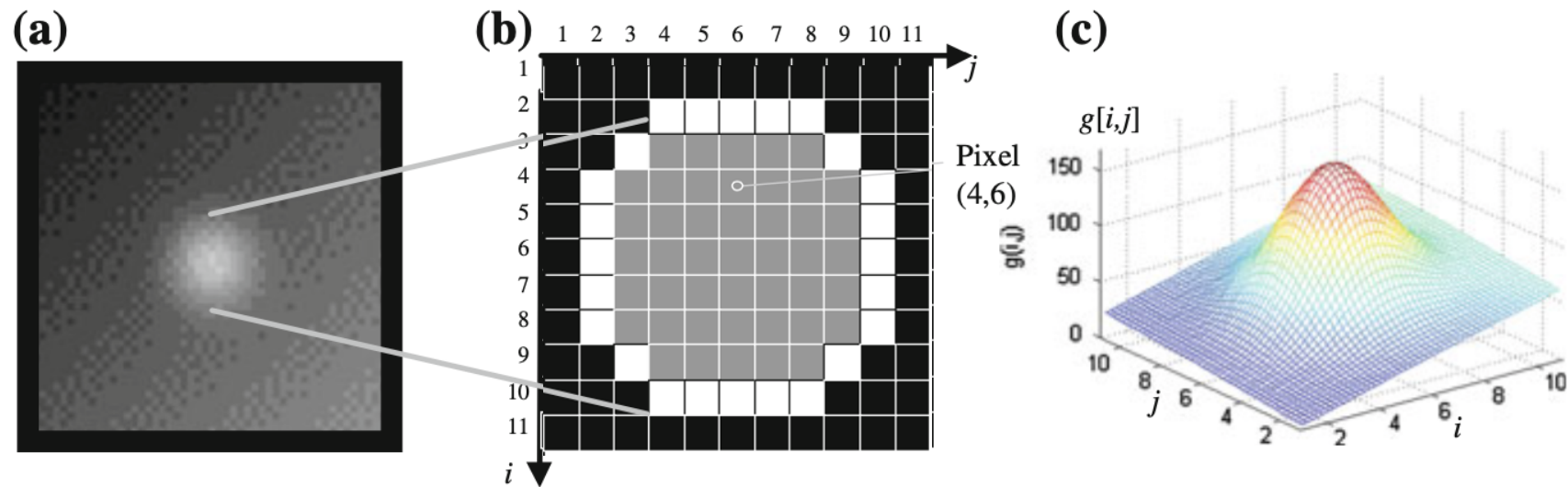
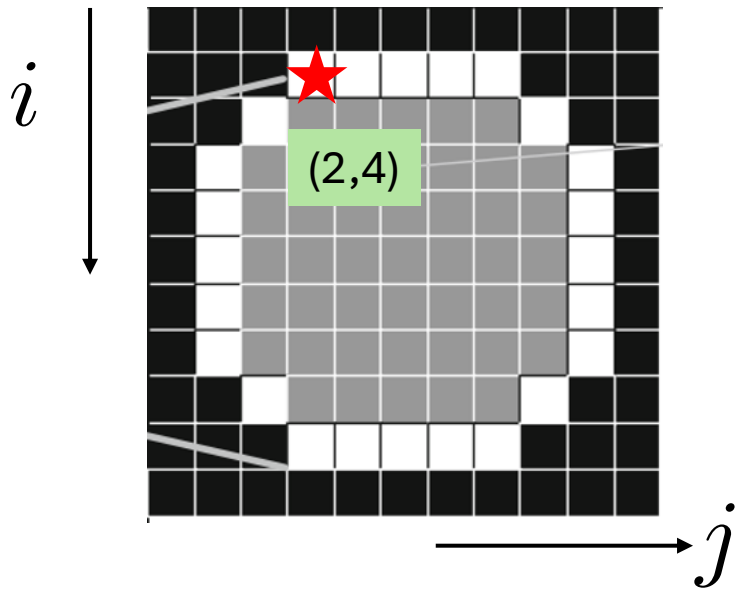
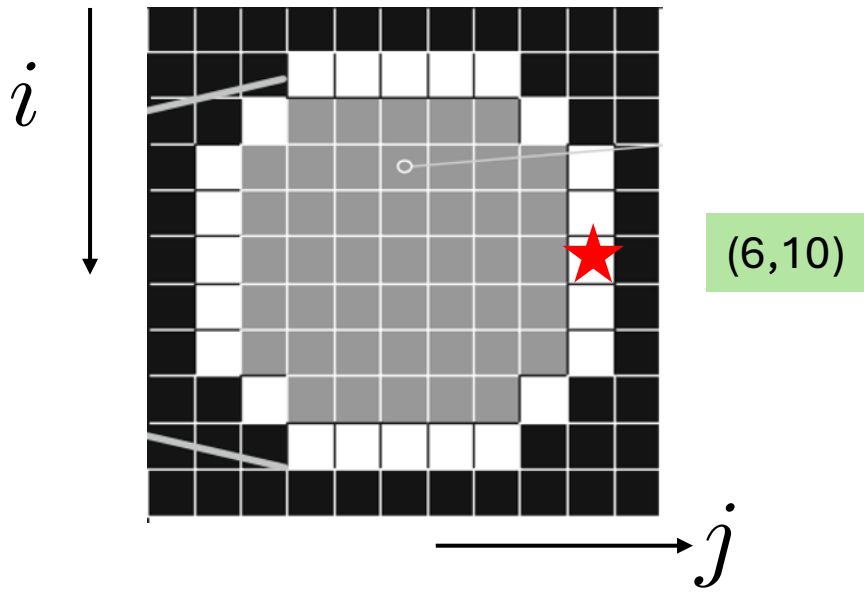
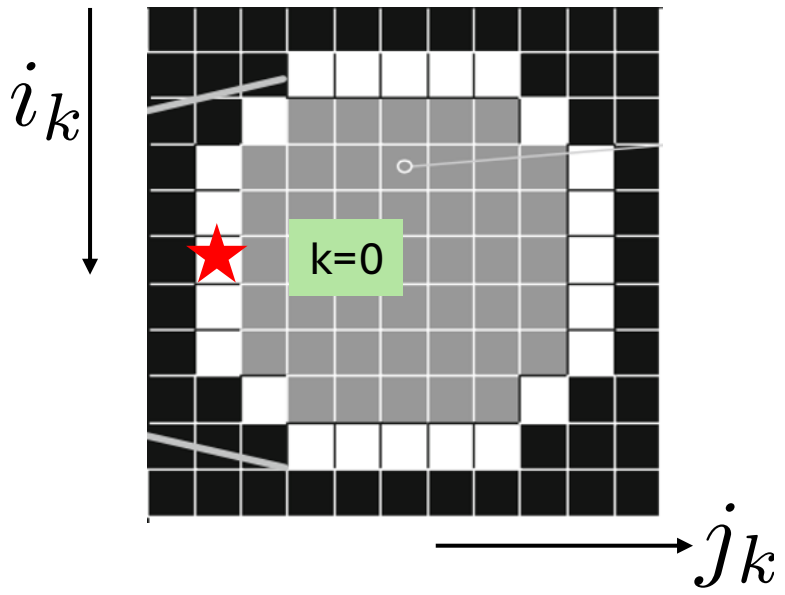
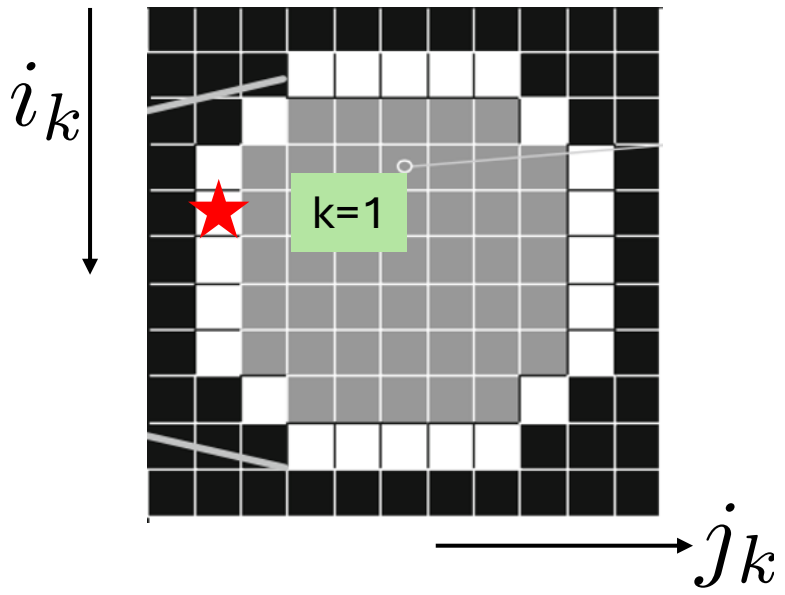


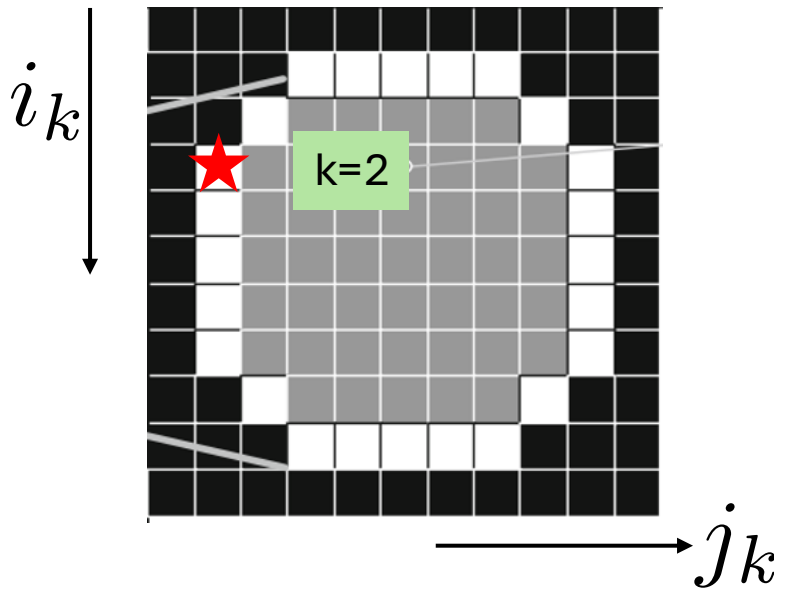
Fig. 5.1 Example of a region: **a** X-ray image, **b** segmented region (*gray* pixels), **c** 3D representation of the gray values

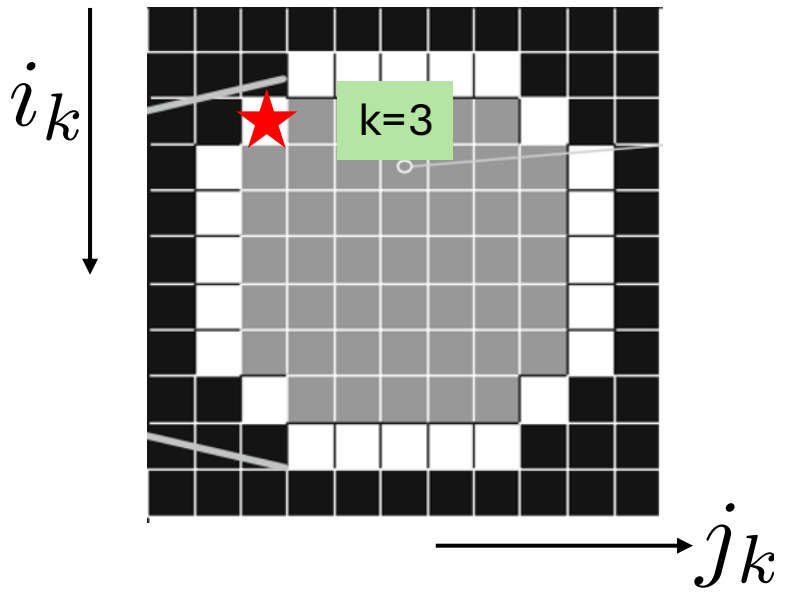


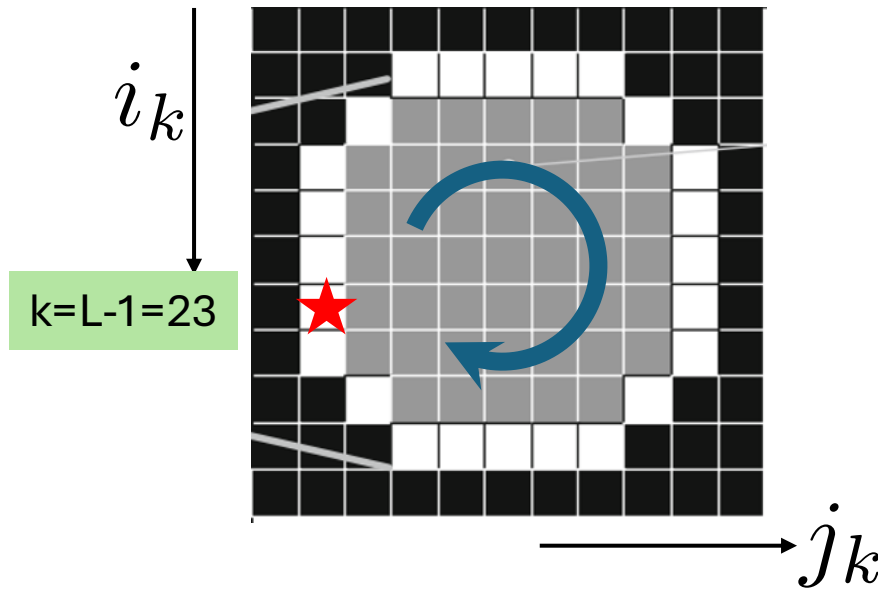


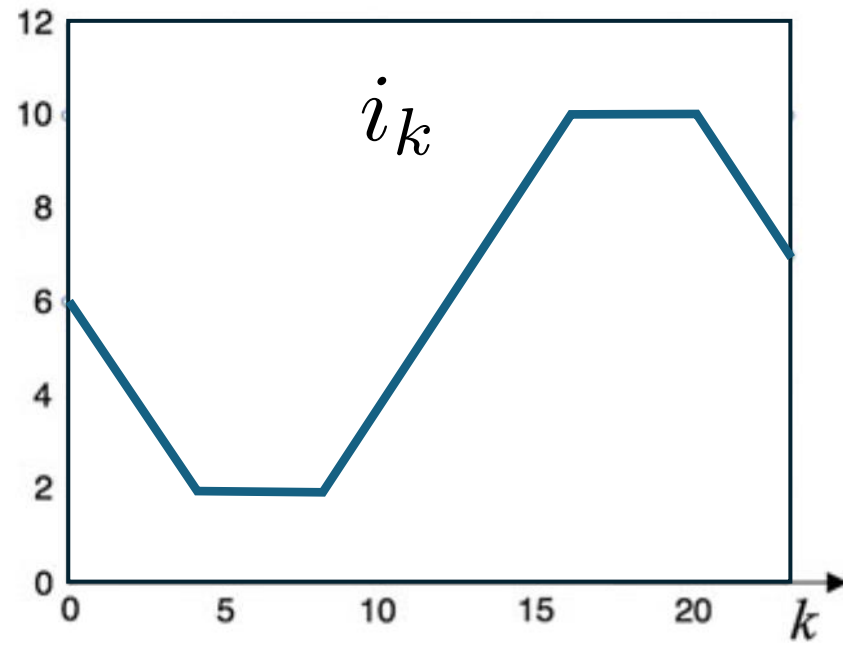
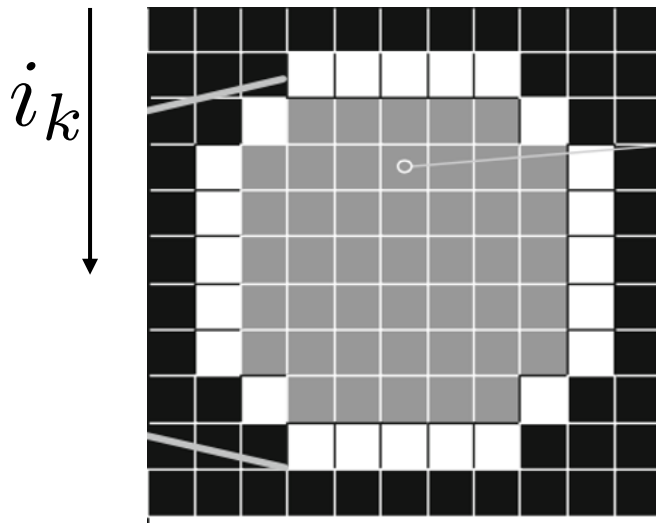


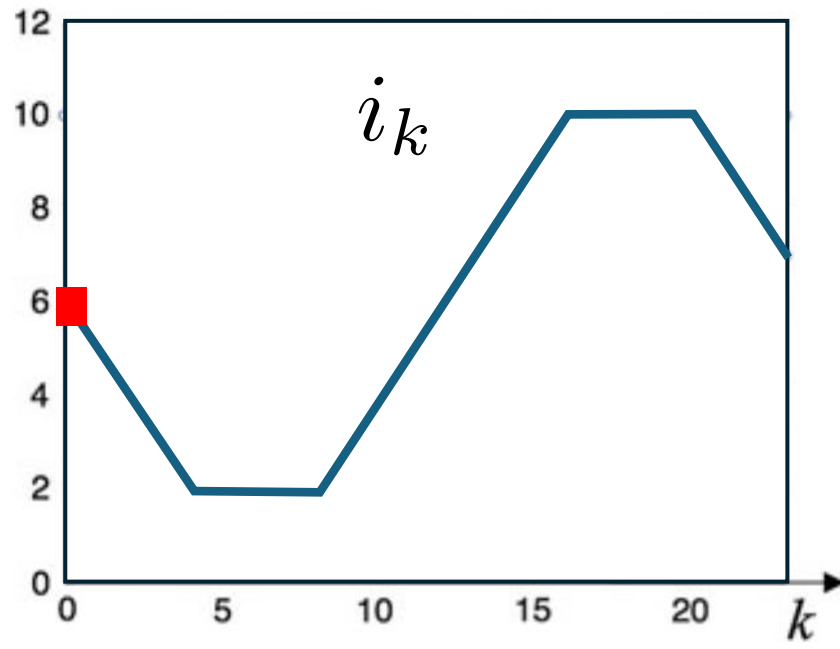
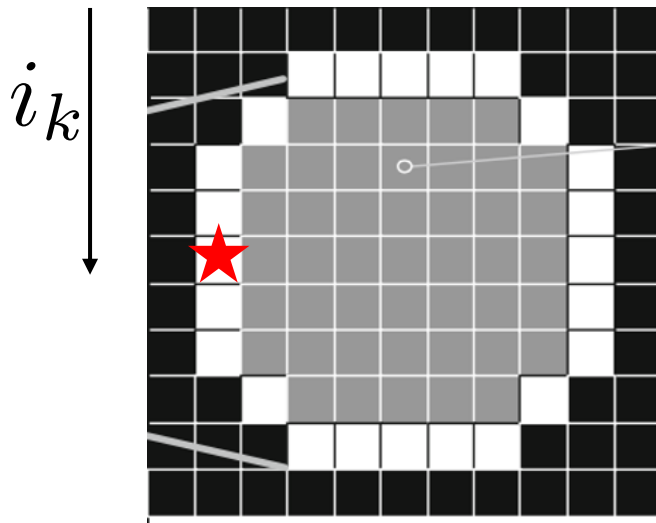


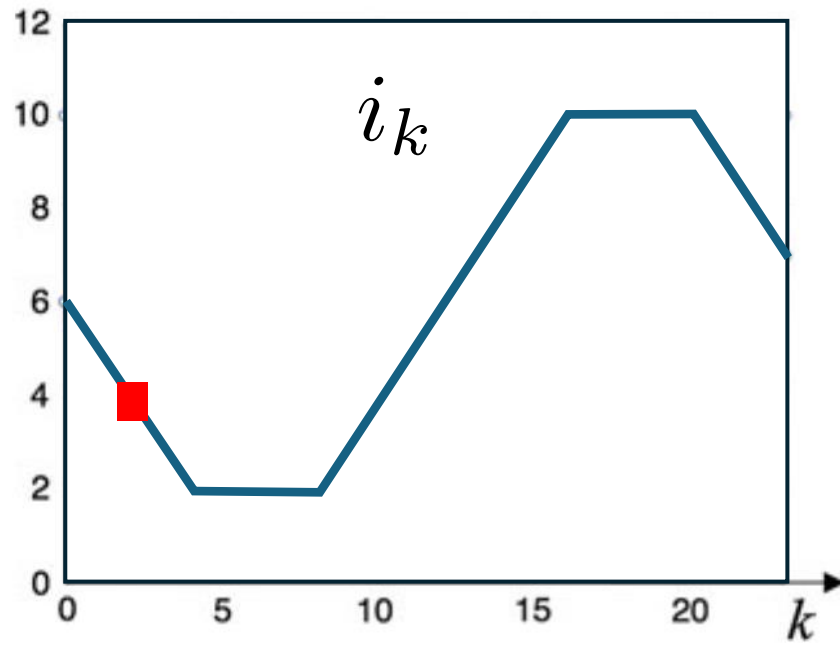
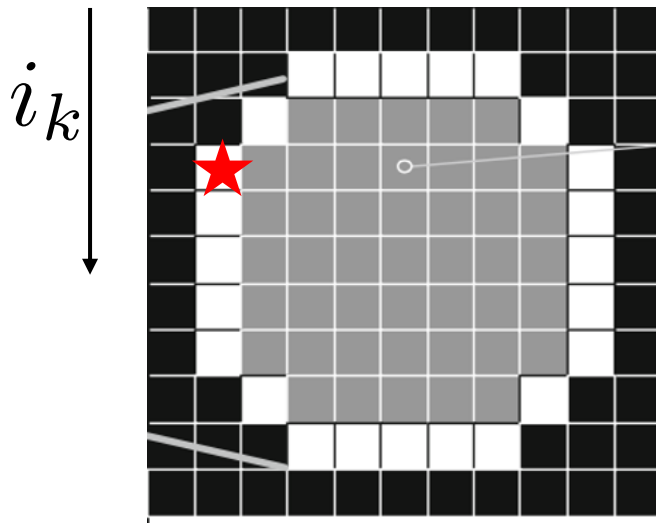


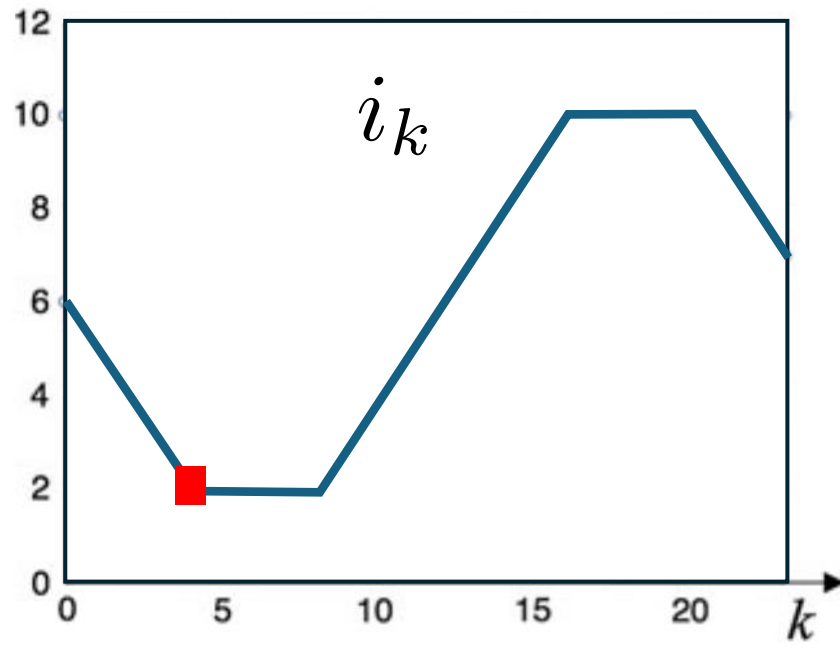
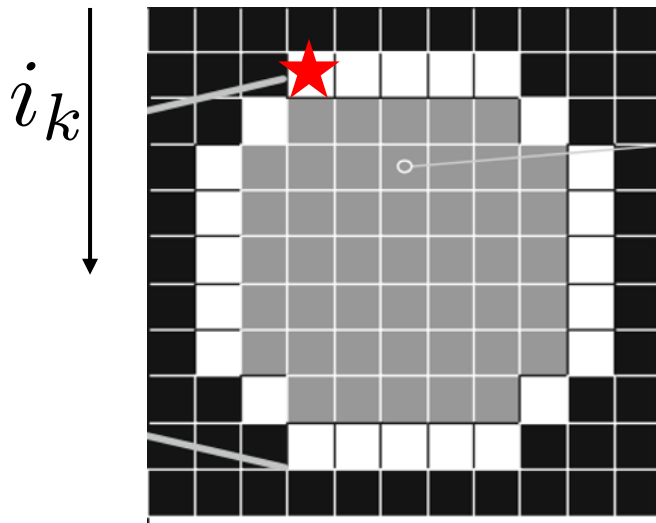


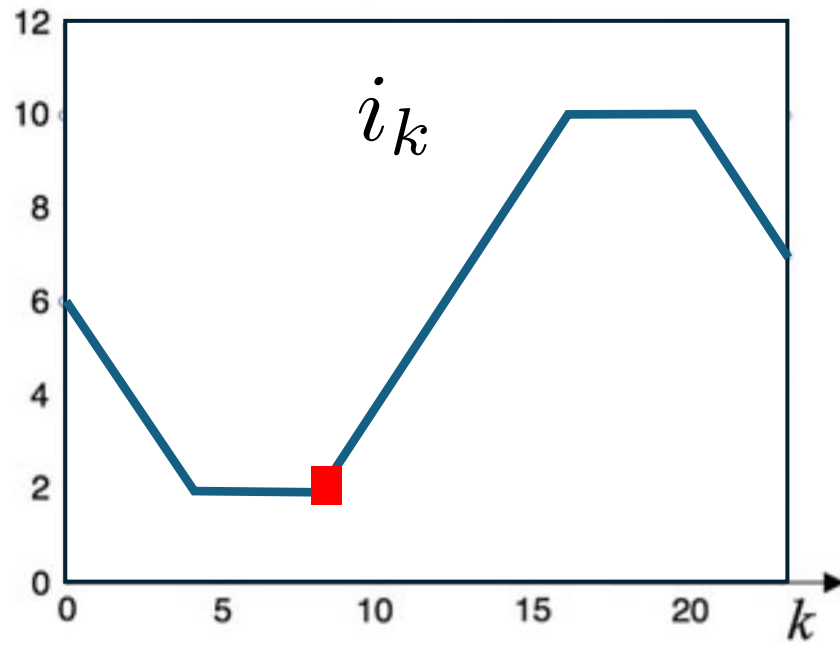
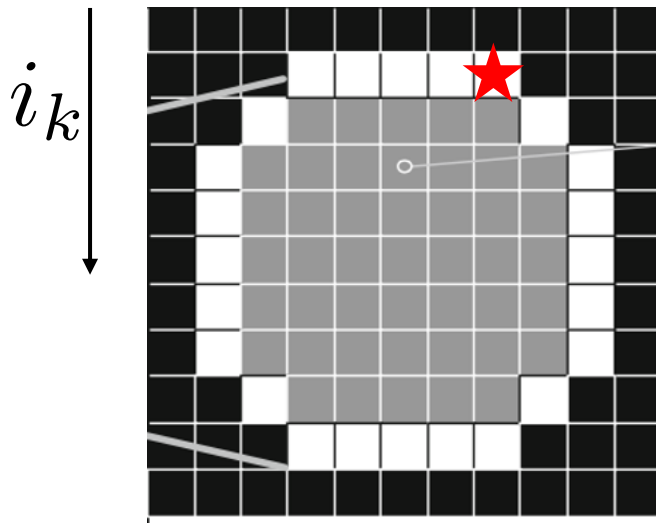


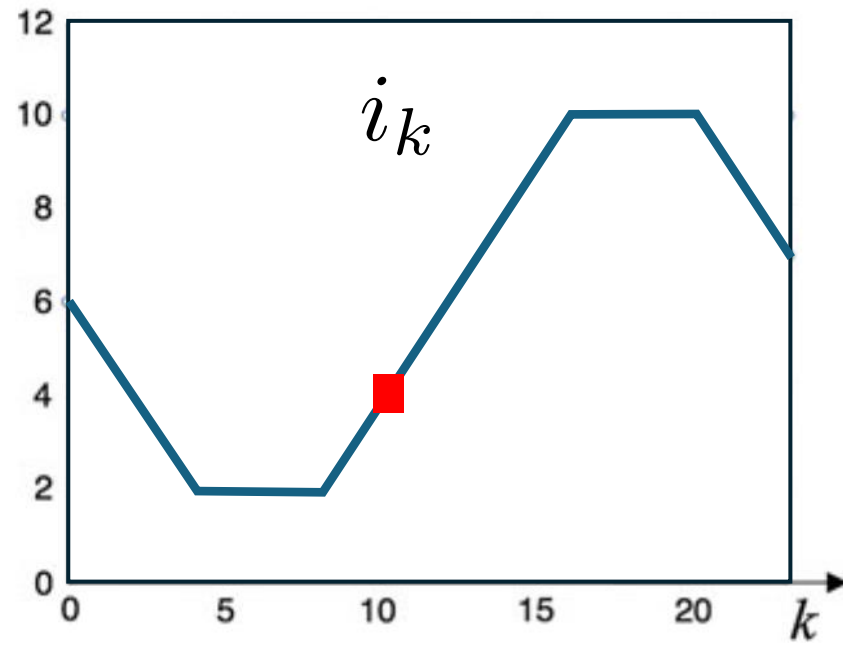
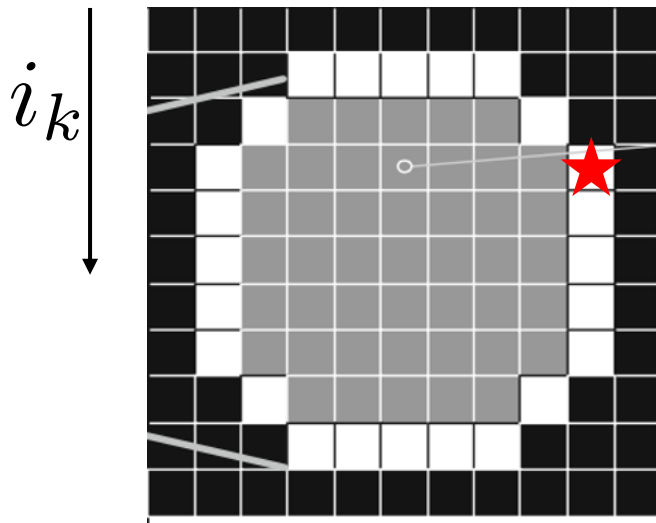


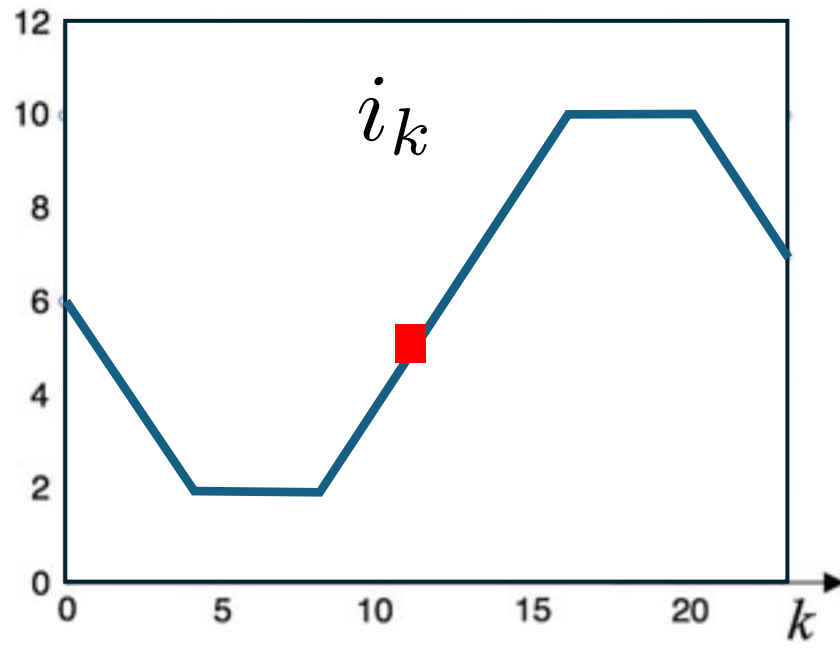
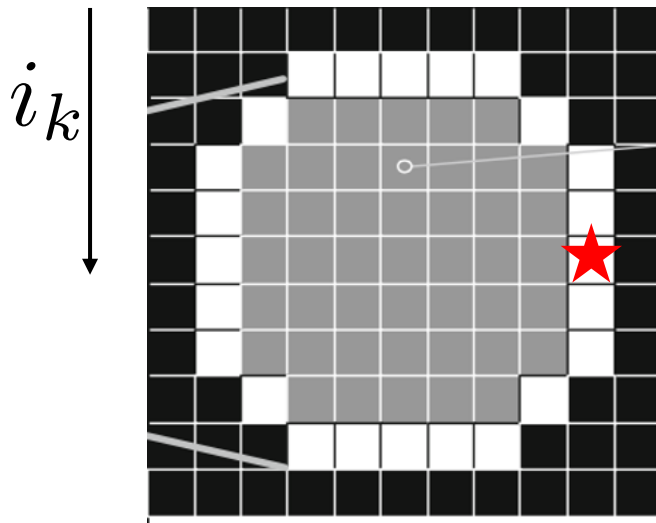


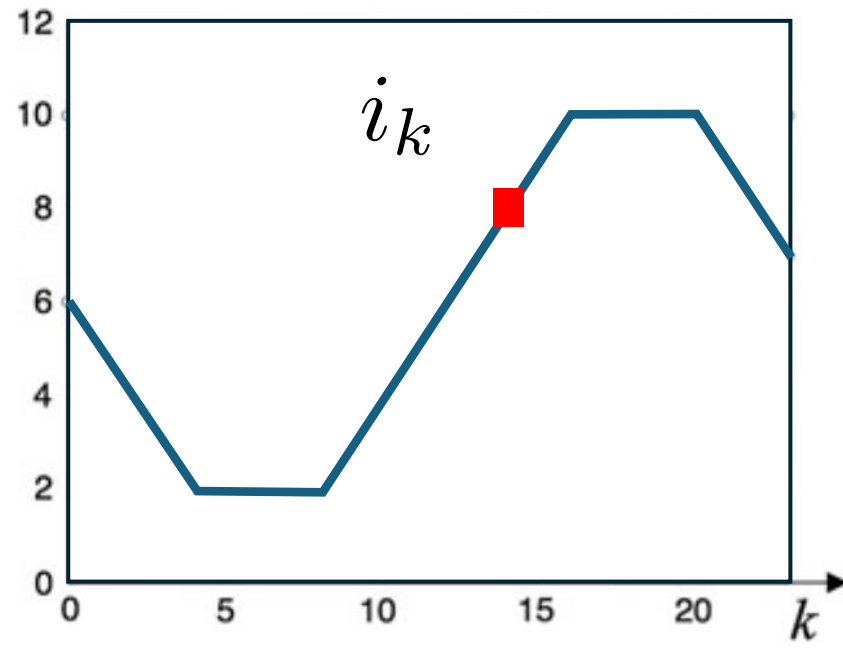
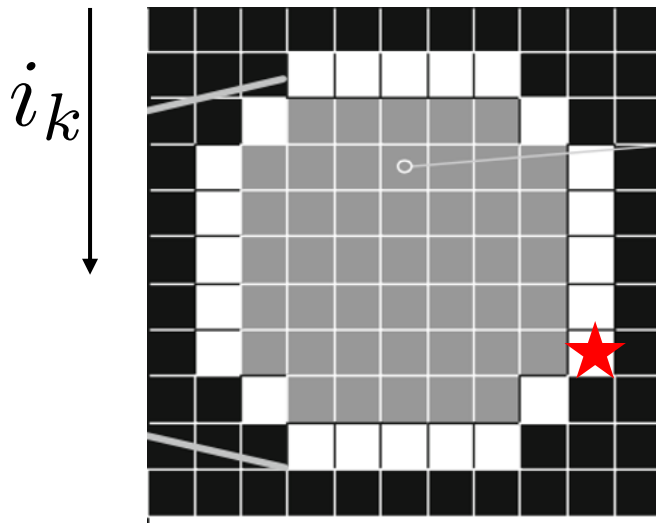


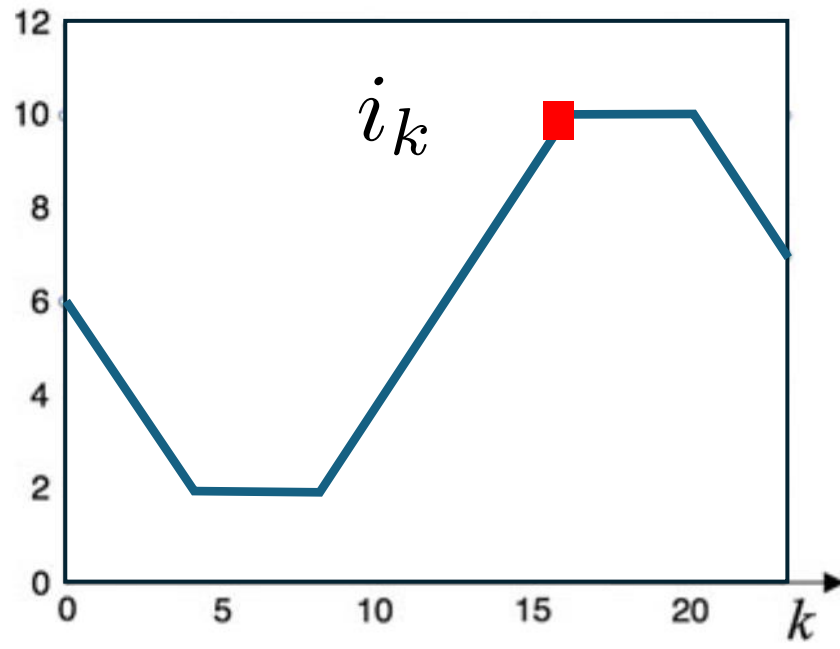
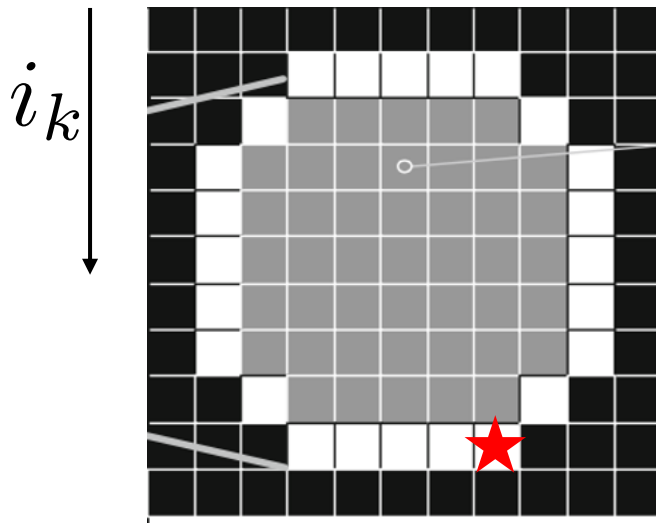


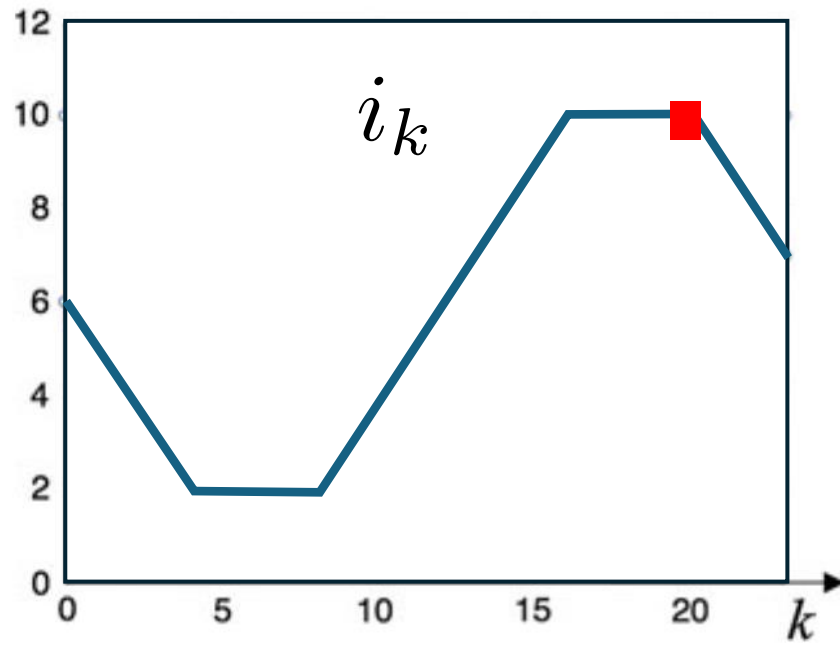
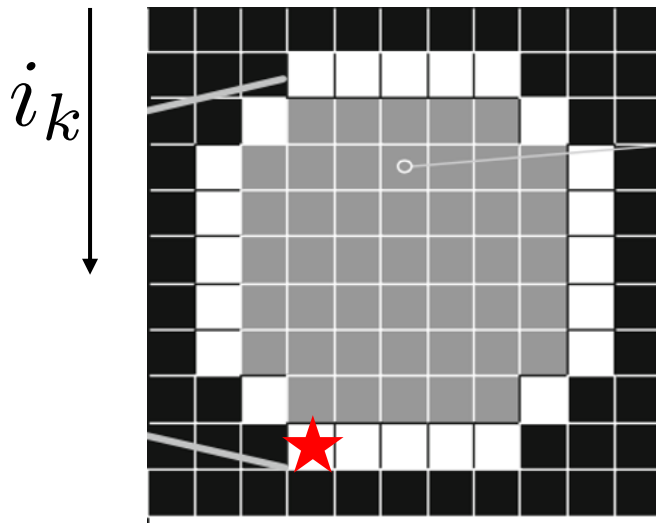


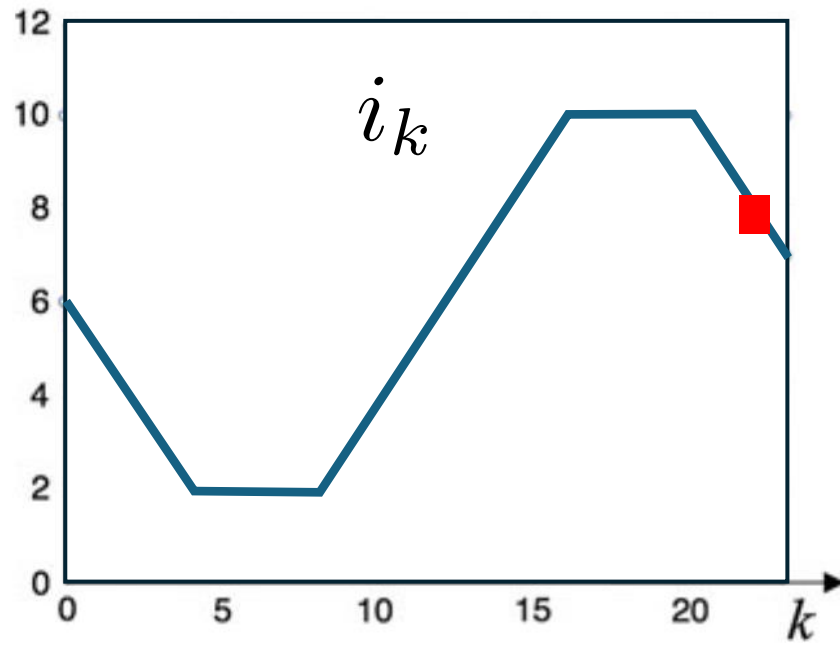
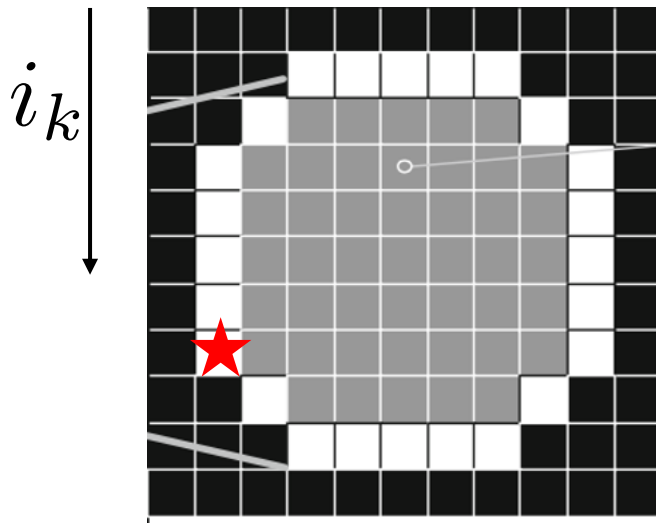


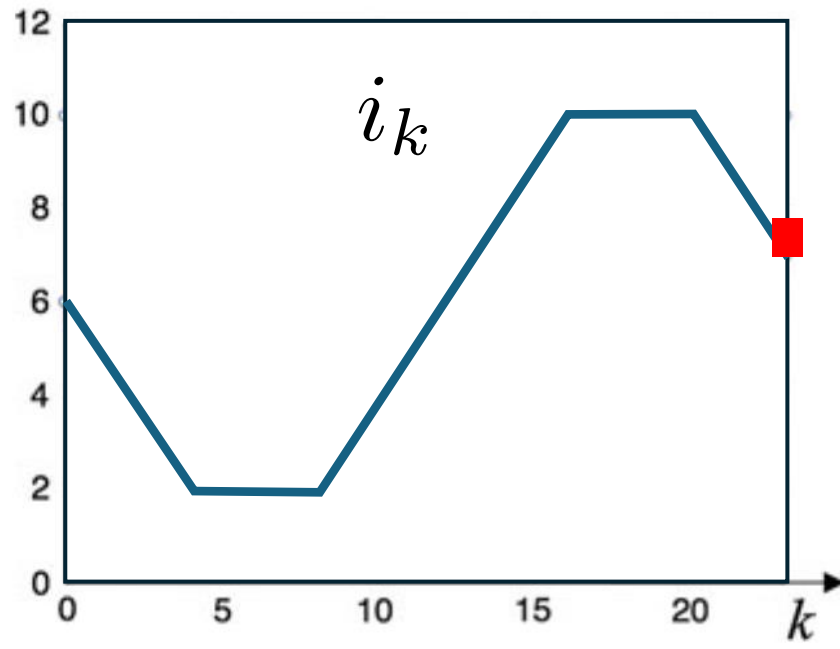
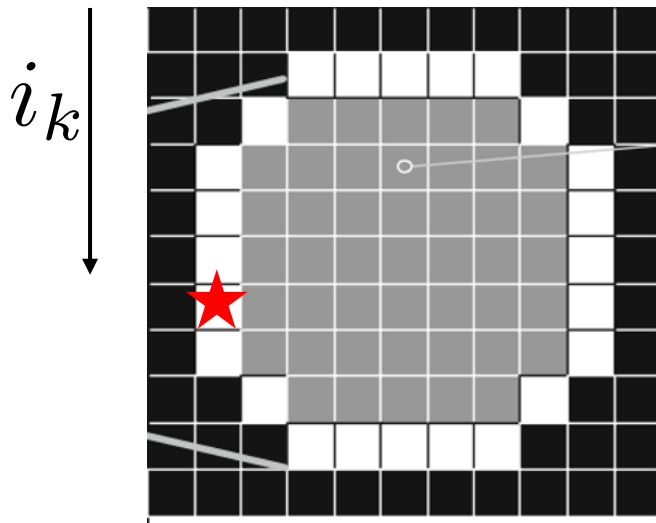


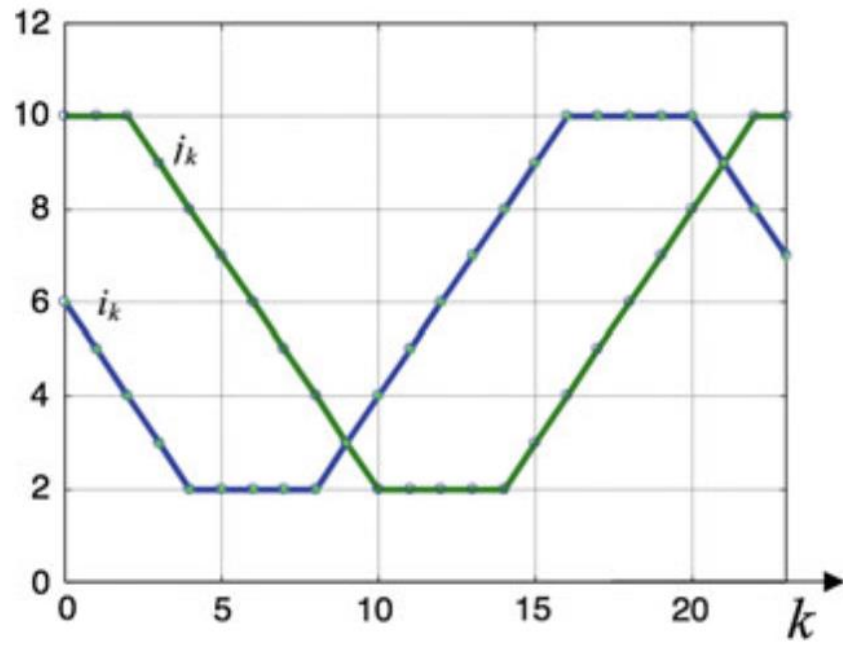
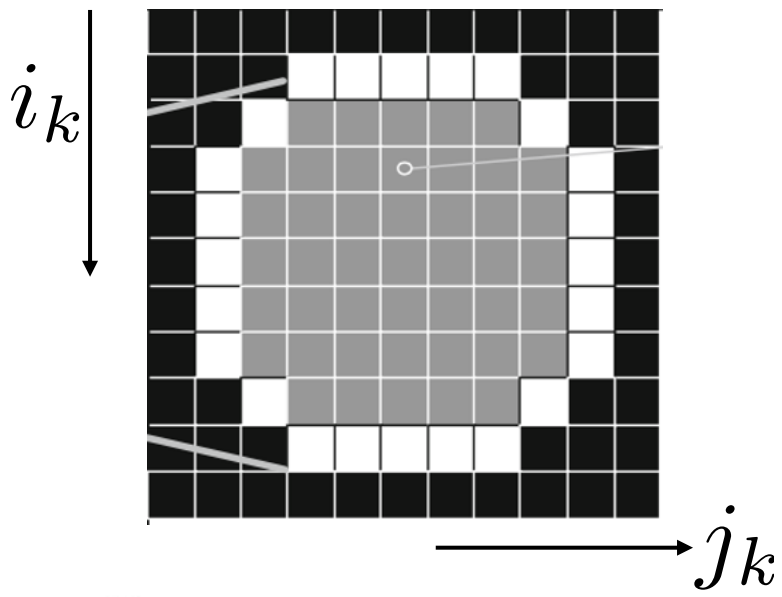


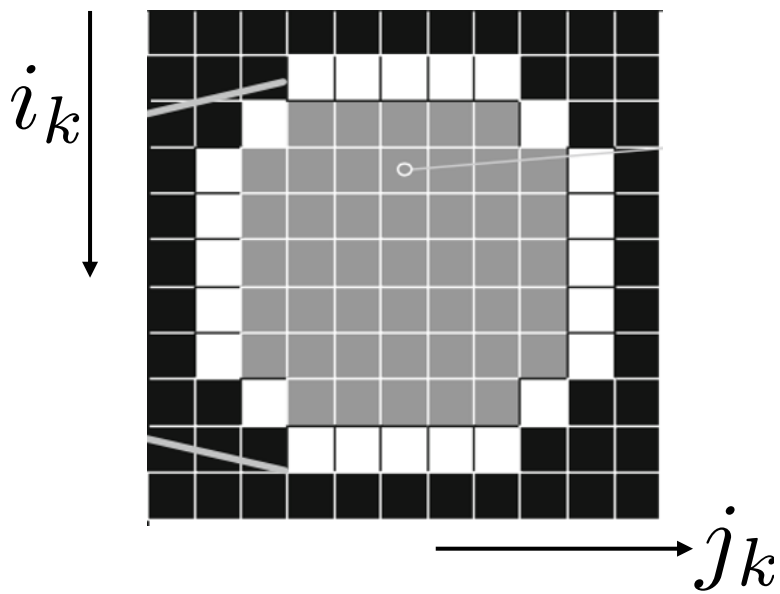












for $n = 0, \dots, L - 1$.

$$F_n = \sum_{k=0}^{L-1} (i_k + j \cdot j_k) e^{-j \frac{2\pi kn}{L}}$$

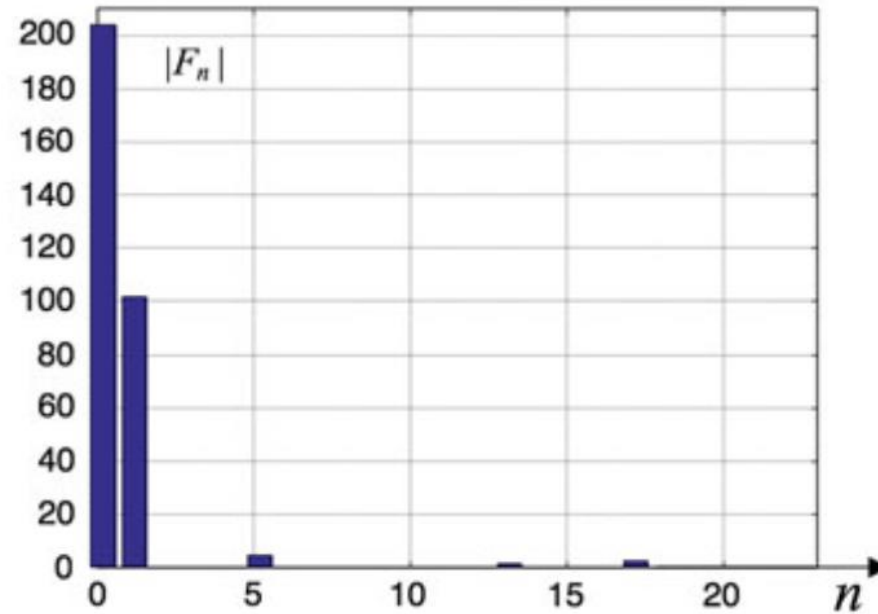
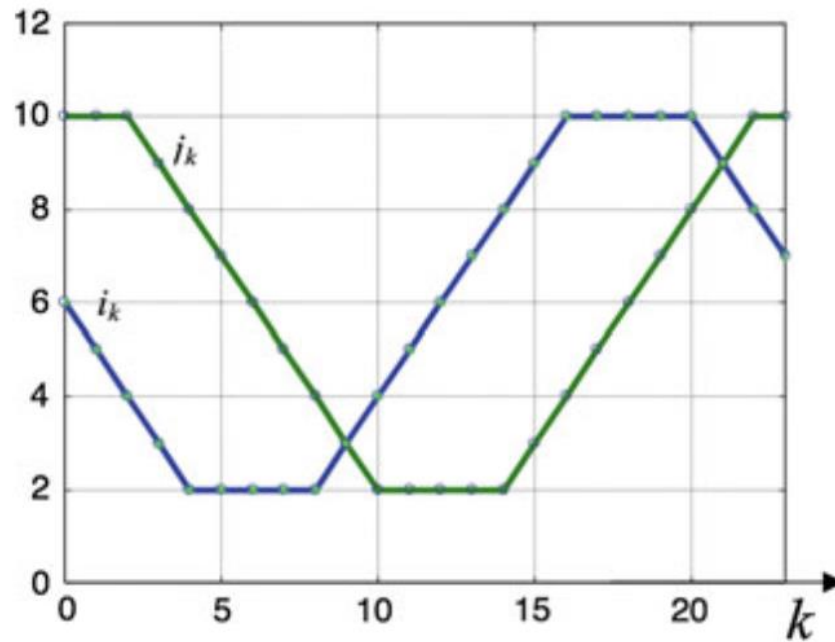


Fig. 5.4 Coordinates of the boundary of region of Fig. 5.1 and the Fourier descriptors

Ejemplo

